WHIZLABS

AZ-305: Azure Solutions Architect Expert

Cheat Sheet

Quick Bytes for you before the exam!

The information provided in Cheat Sheets is for educational purposes only; created in our efforts to help aspirants prepare for the Exam AZ-305 certification. Though references have been taken from Microsoft documentation, it's not intended as a substitute for the official docs. The document can be reused, reproduced, and printed in any form; ensure that appropriate sources are credited and required permissions are received.

Are you Ready for Microsoft Azure "AZ-305" Certification?

Self-assess yourself with

Whizlabs FREE TEST

800+ Hands-on-Labs and Cloud Sandbox

Hands-on Labs **Cloud Sandbox environments**

AZ-305 Cheat Sheet Index

Identity, governance, and monitoring solutions

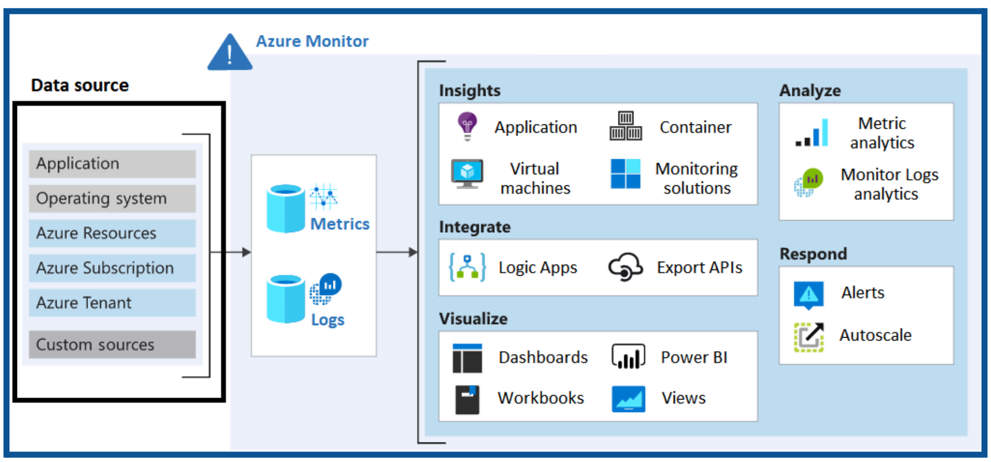
Design solutions for logging and monitoring

Azure Monitor Data Sources

What is Azure Monitor?

- The Azure Monitor serves as your central hub for securing your Azure environment.
- By harnessing its activity log, you gain in-depth visibility into subscription-level events, empowering you to identify potential threats and take swift action.
- This comprehensive log records activities like resource modifications and VM startups, providing valuable context for understanding changes within your Azure infrastructure.
- Access it directly in the portal or leverage PowerShell and the Azure CLI for retrieval.

Azure Monitor is a platform for monitoring and diagnosing issues across applications and infrastructure.



Source: Microsoft Documentation

Azure Monitor Logs (Log Analytics) Workspaces

- → **Azure Monitor Logs (Log Analytics) Workspaces** are unique environments where you can collect and analyze log data from various sources.
  - ◆ Each workspace has its data repository and configuration.
  - ◆ You can combine data from multiple services within a single workspace.
- → It’s like having a dedicated toolbox for monitoring and troubleshooting your cloud and on-premises resources.

Azure Workbooks, insights and Data Explorer

Azure Workbooks

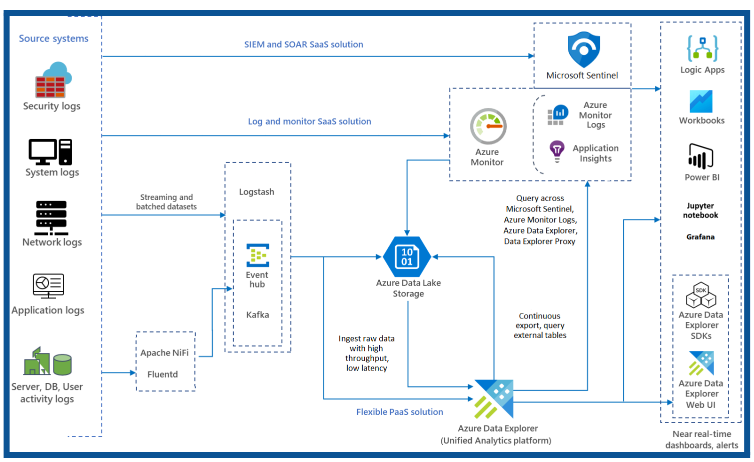
- It provides a flexible canvas for data analysis and the creation of rich visual reports within the Azure portal.
- They allow you to tap into multiple data sources from across Azure and combine them into unified interactive experiences.
- Workbooks combine text, log queries, metrics, and parameters into rich interactive reports.
- They're helpful for scenarios like exploring virtual machine usage, explaining performance, sharing experiment results, and reporting on outages.

Azure Insights

- It is a broader concept, encompassing various monitoring and diagnostic tools in Azure.
- It includes services like Azure Monitor, which provides centralized monitoring for applications and infrastructure, and Azure Application Insights, which focuses on performance monitoring.

Azure Data Explorer

- Azure Data Explorer is a fully managed, high-performance, big data analytics platform that makes it easy to analyze large volumes of data in near real-time.
- It provides an end-to-end data ingestion, query, visualization, and management solution.
- You can collect, store, and analyze diverse data from applications, websites, IoT devices, and more.



Source: Microsoft Documentation

Design authentication and authorization solutions

Identity and access management [IAM]

Identity and Access Management (IAM) is a framework that ensures the right individuals have access to the appropriate resources at the right times for the right reasons.

[Source: Microsoft Documentation]

Identity Management:

This involves creating, storing, and managing identity information. Think of it as keeping track of who's who in your system. Identity providers (IdPs) are like gatekeepers that manage user identities and their associated permissions.

Identity Federation:

Imagine allowing users who already have passwords elsewhere (like in your enterprise network or with an internet or social identity provider) to access your system. It's like extending a friendly handshake across different platforms.

Provisioning and deprovisioning:

This process handles creating and managing user accounts. It includes specifying which users have access to which resources and assigning permissions. When someone leaves the company, their access is "deprovisioned."

Authentication:

This step confirms that a user, machine, or software component is who or what they claim to be. You can add extra security layers like multi-factor authentication (MFA) or use single sign-on (SSO) for convenience.

Authorization:

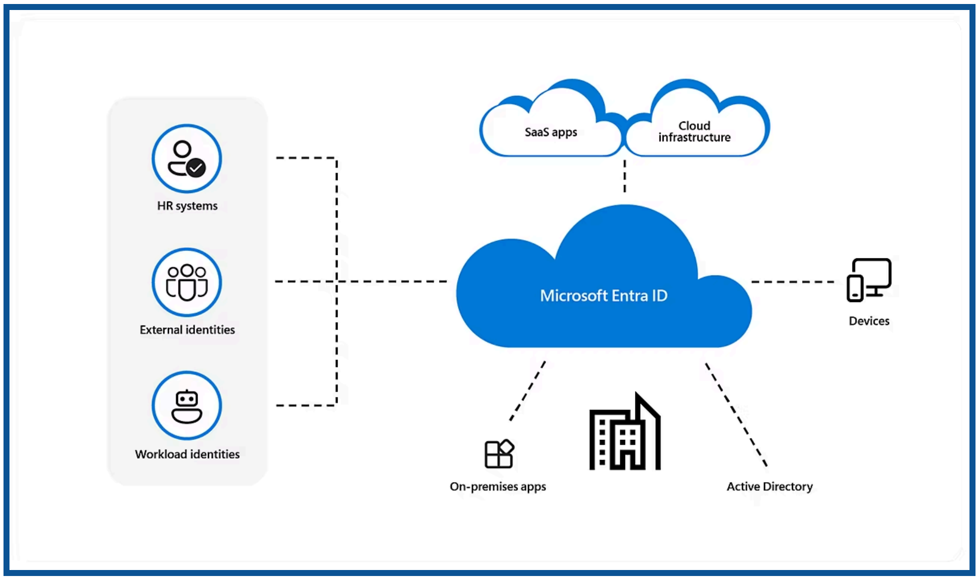
Once authenticated, authorization ensures that users get the exact level and type of access they're entitled to. Users can be grouped into roles, making it easier to manage privileges.

Microsoft Entra Service Overview

What is Microsoft Entra ID?

Microsoft Entra ID is a cloud-based identity and access management solution that lets your employees access external resources like Microsoft 365, the Azure portal, and hundreds of other SaaS applications.

- It’s a suite of tools of identity and access management (IAM) solutions from Microsoft.
- Microsoft's cloud-based identity and access management service.
- It helps organizations manage and secure access to their applications, data, and resources.



[Source: Microsoft Documentation]

Microsoft Entra Capabilities:

- ➔ App integrations and single sign-on (SSO)
- ➔ Passwordless and multifactor authentication (MFA)
- ➔ Conditional access and Identity protection
- ➔ Privileged identity management
- ➔ End-user self-service, and Unified admin center

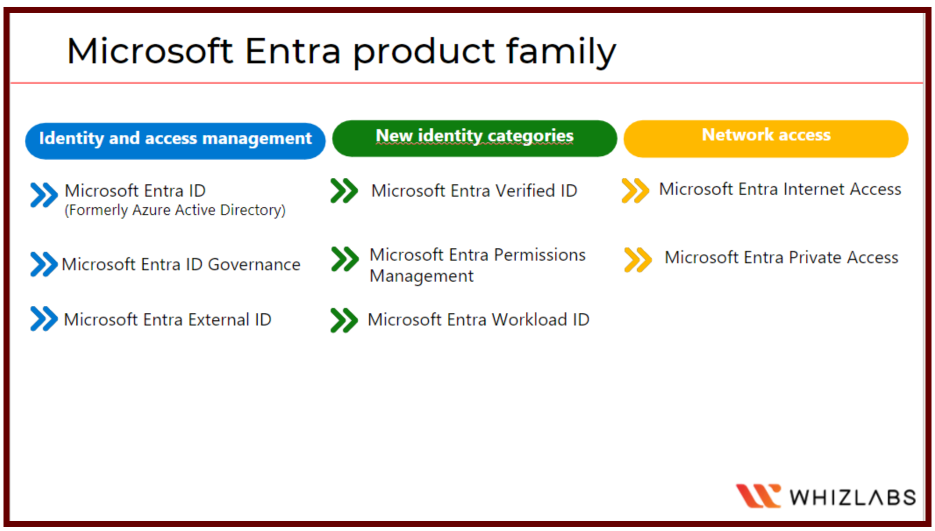
Microsoft Entra ID Licenses

- **Microsoft Entra ID Free:** Offers self-service password changes for cloud users, group and user management, basic reporting, on-premises directory synchronization, and single sign-on for Azure, Microsoft 365, and numerous well-known SaaS apps.

- **Microsoft Entra ID P1:** P1 provides hybrid users with access to both on-premises and cloud resources, in addition to the free features. It also supports advanced administration features, including dynamic groups, self-service group management, Microsoft Identity Manager, and cloud write-back capabilities that enable on-premises users to reset their passwords themselves.

- **Microsoft Entra ID P2:** In addition to the Free and P1 features, P2 includes Microsoft Entra ID Protection, which helps provide risk-based Conditional Access to your apps and critical company data, and Privileged Identity Management, which helps discover, restrict, and monitor administrators and their access to resources, as well as provide just-in-time access when necessary.

- **"Pay as you go" feature licenses:** Additionally, licenses for features like Microsoft Entra Business-to-Customer (B2C) are available. You can give your customer-facing apps identity and access management solutions with the assistance of B2C.



Secure Microsoft Entra users

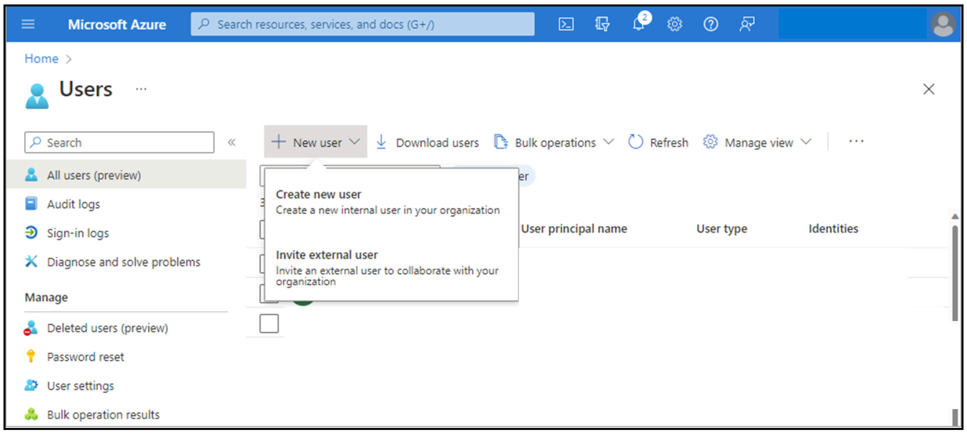
- We can add new users or delete existing users from Microsoft Entra ID Tenant.
- Only the User Administrator or Global Administrator is authorized to add or delete users

How to add a new user?

We can create a new user for your organization or invite an external user.

1. Sign in to the Azure portal with the user administrator role.
2. Microsoft Entra ID > Users.

3. Select Create New User or Invite External User from the menu.



[Source: Microsoft Documentation]

4. New User page ⟶ provide the new user's information:
5. Copy the auto generated password provided in the Password box. You must provide this password to the user to sign in for the first time.
6. Then Select Create.

The user is created and added to your Microsoft Entra ID organization.

MS Source: [Secure users in Microsoft Entra ID - Training | Microsoft Learn]

Add a new guest user

You can also invite a new guest user to collaborate with your organization by selecting the Invite User option from the New User page. If your organization's external collaboration settings are configured to allow guests, an invitation is emailed that the user must accept to start collaboration.

Add other users

There may be times when you want to manually create user accounts in your Microsoft Entra B2B directory. If you have an environment with Microsoft Entra ID (cloud) and Windows Server Active Directory (on-premises), you can add new users by synchronizing existing user account data.

Delete the users: **We can delete an existing user using the Microsoft Entra ID portal.**

- You must have a Global Administrator, Privileged Authentication Administrator, or User Administrator role assignment to delete users in your organization.

- Global Admins and Privileged Authentication Admins can delete any users, including other admins.
- User Administrators can delete any non-admin users, Helpdesk Administrators, and other User Administrators.

1. Sign in to the **Azure portal** using one of the appropriate roles listed above.
2. Go to **Microsoft Entra ID** > **Users**.
3. Search for and select the user you want to delete from your Microsoft Entra ID tenant.
4. Select **Delete user**.

[Source: Microsoft Documentation]

Secure Microsoft Entra groups

What are Microsoft Entra groups?

Microsoft Entra groups manage users who all need the same access and permissions to resources, such as potentially restricted apps and services. Instead of adding special permissions to individual users, you create a group that applies the special permissions to every group member.

Using groups also enables the following management features:

- Attribute-based dynamic groups
- External groups synced from on-premises Active Directory
- Administrator-managed or self-service-managed groups

Microsoft Entra ID allows you to use groups to control access to applications, data, and resources. The resources can be:

- Microsoft Entra ID that grants access to manage objects
- Outside the company, like in the case of Software as a Service (SaaS) applications
- Azure Services
- SharePoint websites
- On-premises resources.

The Azure portal is unable to manage the following groups:

- Only on-premises Active Directory is used to manage groups synchronized with it.

- The only places to manage distribution lists and mail-enabled security groups are the Microsoft 365 admin center and the Exchange admin center. To manage these groups, you need to log in to the Microsoft 365 admin center or the Exchange admin center.

Types of groups:

Security: Controls how computers and users access shared resources. To ensure that each member of the group has the same set of security permissions, for instance, you can create a security group. Users, devices, service principals, and other groups (also called nested groups) that specify permissions and access policies can all be members of a security group. Users and service principals are examples of who can own a security group.

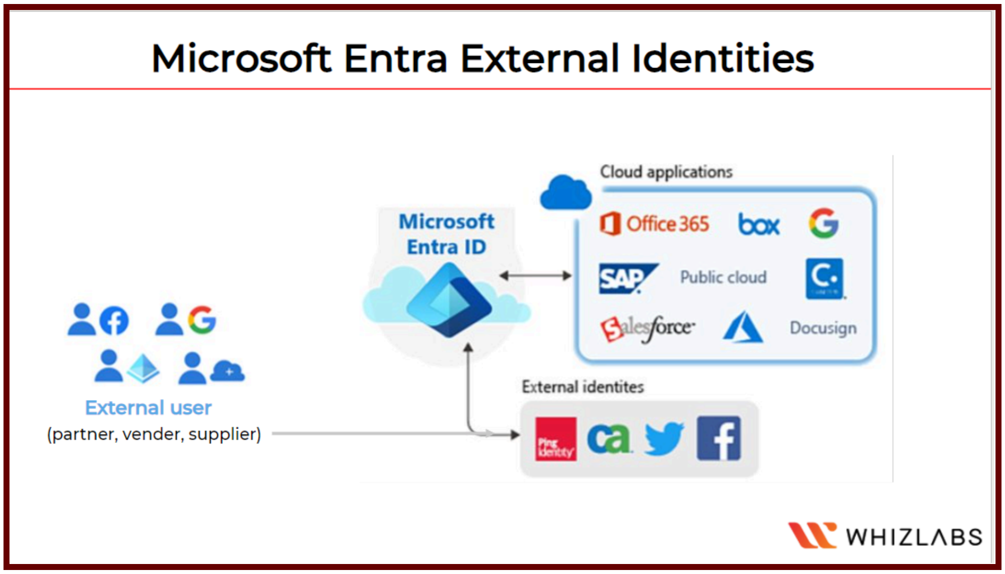
Microsoft 365: Members of a group can collaborate by using Microsoft 365, which grants access to a shared calendar, files, mailbox, SharePoint sites, and more.

Types of membership:

- **Assigned:** This allows you to give particular users special permissions to join a group.
- **Dynamic user:** This allows you to automatically add and remove members based on dynamic membership rules. When a member's attributes change, the system examines your dynamic group rules for the directory to determine whether the member is added or removed based on whether they still meet the requirements of the rule.
- **Dynamic device:** This enables automatic device addition and removal using dynamic group rules. When a device's attributes change, the system checks your dynamic group rules for the directory to determine whether the device should be added to the rule or removed based on whether it still meets the requirements.

Microsoft Entra External Identities

The term "Microsoft Entra External ID" describes all of the secure methods through which you can communicate with users outside of your company. If you want to collaborate with distributors, suppliers, vendors, or partners, you can define resource sharing and how internal users can access external organizations. As a developer building apps for users, you can control the identity experiences of your clients.



External users can "bring their own identities" by using External ID. They can use their credentials to log in regardless of whether they have an unmanaged social identity like Google or Facebook or a digital identity issued by a company or government.
External identities are made up of the following capabilities:

B2B collaboration: Allow external users to sign in to your Microsoft or other enterprise applications (SaaS apps, custom-developed apps, etc.) using their preferred identity. This will allow you to collaborate with them.
B2B direct connect - Create a mutual, two-way trust with another Microsoft Entra organization to enable seamless collaboration. B2B direct connect currently supports Teams shared channels, which allow external users to access your resources from within their own Teams instances.

B2C - Use this B2C for identity and access management when publishing contemporary SaaS apps or custom-developed apps (apart from Microsoft apps) to users and clients.

Microsoft Entra multi-tenant organization - Collaborate with multiple tenants in a single Microsoft Entra organization using cross-tenant synchronization.

Secure external identities

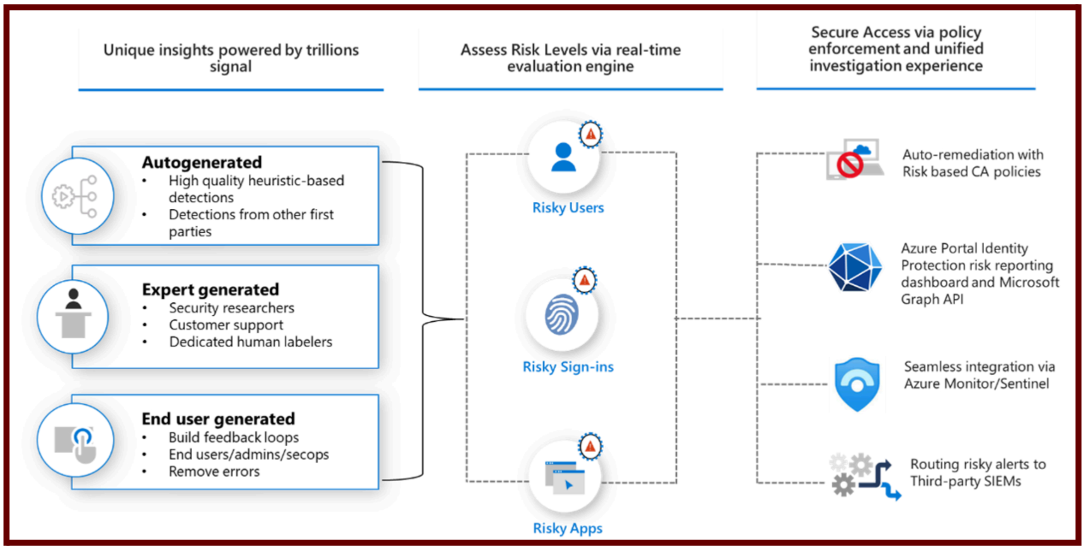
When to Use External Identities:

- External Identities are especially useful in situations where collaboration extends beyond the organization's boundaries.
- If you want to collaborate with entities like partners or developers to create consumer-facing applications, External Identities are a strategic solution.

- It provides a framework for users from other organizations to bring their own identities, whether they have a corporate, government-issued digital identity or use unmanaged social identities such as those from Google and Facebook.

Implement Microsoft Entra ID Protection

Microsoft Entra ID Protection is a tool that enables organizations to automate the detection and remediation of identity-based risks, conduct risk investigations using portal data, and export risk detection results to third-party utilities for further analysis.



[Source: Microsoft Documentation]

Microsoft Entra ID Protection is a tool that allows organizations to accomplish three key tasks:

- Automate the detection and remediation of identity-based risks.
- Investigate risks using data in the portal.
- Export risk detection data to third-party utilities for further analysis.

★ Microsoft Entra ID Protection enables businesses to detect, investigate, and mitigate identity-based risks.

★ These identity-based risks can then be fed into tools such as Conditional Access to make access decisions or back into a security information and event management (SIEM) tool for further investigation and correlation.

Types of Risks:

- Anonymous IP address usage
- Password spray attacks
- Leaked credentials

Investigate:

Any risks detected on an identity are tracked with reporting. Identity Protection provides three key reports for administrators to investigate risks and take action:

- ➔ Risk detections: Each risk detected is reported as a risk detection.
- ➔ Risky sign-ins: A risky sign-in is reported when there are one or more risk detections reported for that sign-in.
- ➔ Risky users: A Risky user is reported when either or both of the following are true:
    - ◆ The user has one or more Risky sign-ins.
    - ◆ One or more risk detections have been reported.

Manage Microsoft Entra Authentication

Microsoft Entra authentication includes the following components:

- Self-service password reset
- Microsoft Entra multifactor authentication
- Hybrid integration to write password changes back to on-premises environment
- Hybrid integration to enforce password protection policies for an on-premises environment
- Passwordless authentication

1. Self-service password reset

Self-service password reset enables users to change or reset their passwords without the involvement of an administrator or help desk. If a user's account is locked or they forget their password, they can use the prompts to unblock themselves and return to work. This ability reduces help desk calls and productivity loss when a user is unable to sign in to their device or application.

Self-service password reset works in the following scenarios:

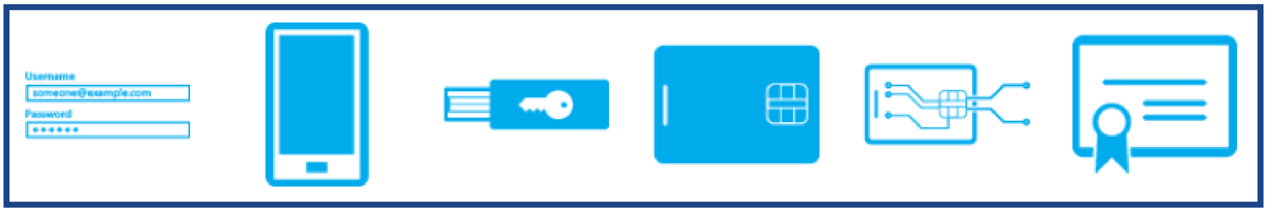
- Password change occurs when a user knows their password but wishes to change it to something new.
- Password reset occurs when a user is unable to sign in, such as when they have forgotten their password and wish to reset it.
- Account unlocking occurs when a user is unable to sign in because their account has been locked and wishes to unlock their account.

A self-service password reset also writes the password back to an on-premises Active Directory when a user changes or resets their password.

Ensuring that a user can use their updated credentials with on-premises devices and applications right away is ensured by password writeback.

2. Microsoft Entra multifactor authentication

Multifactor authentication is a sign-in process in which a user is prompted for an additional form of identification, such as entering a code on their phone or providing a fingerprint scan.



Microsoft Entra multifactor authentication uses a few authentication methods [as follows]

- Typically a password [Something you know]
- Trusted device that is not easily duplicated, like a phone or key [Something you have]
- Biometrics like a fingerprint or face scan [Something you are]

Users can register for both self-service password reset and Microsoft Entra multifactor authentication in a single step, making the onboarding process easier. Administrators can specify which types of secondary authentication can be used. To further secure the self-service password reset process, users may be required to use Microsoft Entra multifactor authentication.

3. Password protection

Microsoft Entra ID by default prevents weak passwords like Password1. A global list of prohibited passwords that are enforced and updated automatically contains known weak passwords. If a Microsoft Entra user attempts to set their password to one of these weak passwords, they are prompted to select a more secure password.
You can set up custom password protection policies to improve security.

4. Passwordless authentication

Eliminating the need for passwords during sign-in procedures is the ultimate objective for numerous settings. A username and password are still weak forms of authentication that can be exposed or brute-forced but features like Azure password protection or Microsoft Entra multi-factor authentication help improve security.

Credentials are supplied using techniques like biometrics with Windows Hello for Business or a FIDO2 security key when you log in without a password. An attacker cannot easily replicate these authentication methods.

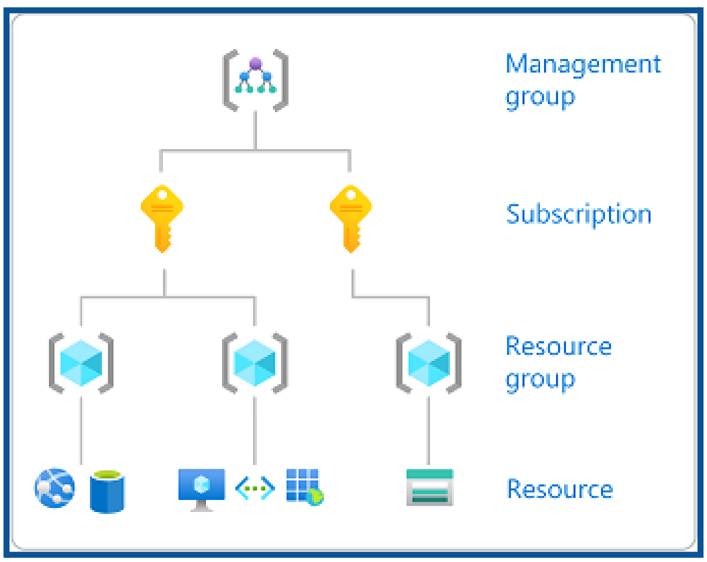
Microsoft Entra ID makes it easier for users to sign in and lowers the risk of attacks by offering native authentication through passwordless methods.

Microsoft Entra Authorization

Azure Role-Based Access Control (RBAC):

Azure RBAC is a fundamental concept that governs access to Azure resources. It operates based on assigning roles to entities (users, groups, or applications) at various levels within the Azure hierarchy.

Scopes in Azure RBAC:

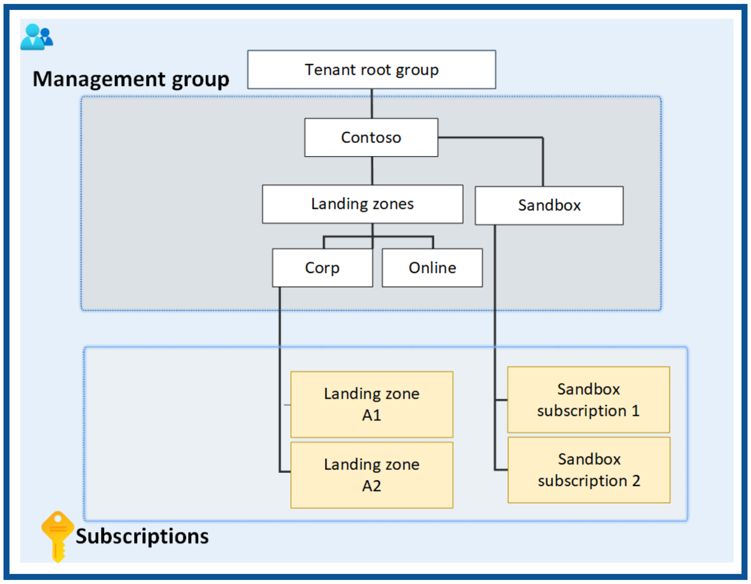


[Source: Microsoft Documentation]

RBAC operates at different scopes, each serving a distinct purpose:

1. **Management Groups:** Management groups provide a way to manage access, policies, and compliance across multiple subscriptions. Configuring role permissions at this level ensures consistent governance throughout your Azure hierarchy.
2. **Subscriptions:** Subscriptions represent agreements with Microsoft to use Azure services. Role assignments at the subscription level help delegate responsibilities and control actions within a specific subscription.

3. **Resource Groups:** Resource groups are logical containers for resources deployed within a subscription. Configuring role permissions at the resource group level allows granular control over access to specific sets of resources.



[Source: Microsoft Documentation]

4. **Resources:** Resources include various Azure services like virtual machines, databases, or storage accounts. Configuring role permissions at the resource level provides fine-tuned control over who can interact with a particular resource.

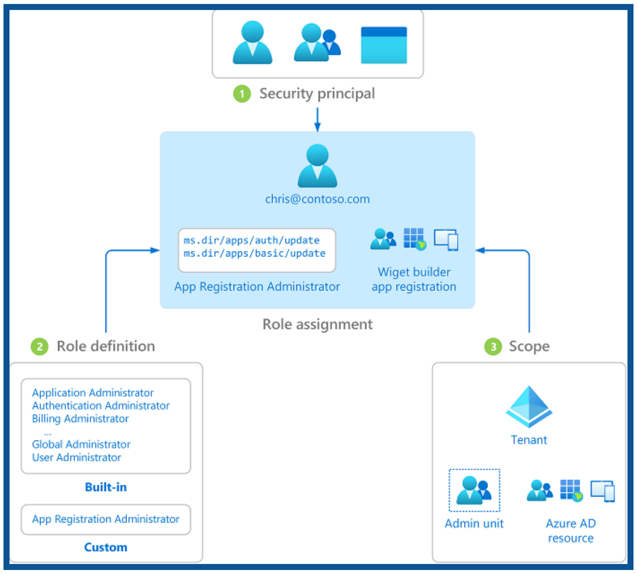
RBAC Roles:

Azure provides several built-in roles catering to different levels of access.
Below are some key roles that included in RBAC:

- **Owner:** Have full access to all resources and can delegate access to others.
- **Contributor:** can create & manage all types of Azure resources, but can't grant access to others.
- **Reader:** Readers can view resources but cannot make any changes.
- **Custom Roles:** Azure allows the creation of custom roles to meet specific business needs, providing flexibility in access control.

Assign Microsoft Entra built-in roles

Microsoft Entra ID supports two types of role definitions: Built-in roles & Custom roles

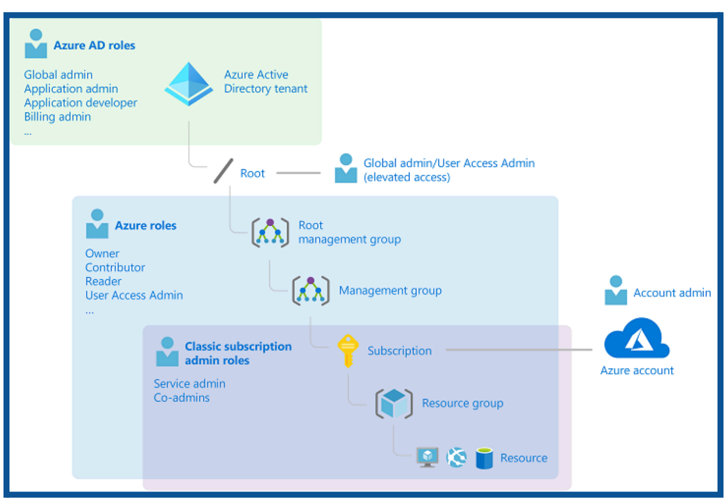


[Source: Microsoft Documentation]

Microsoft Entra roles are used to manage Microsoft Entra resources in a directory, such as creating or editing users, delegating administrative roles, resetting user passwords, managing user licenses, and managing domains.

Assign Azure built-in roles

You can assign various Azure built-in roles to users, groups, service principals, and managed identities through Azure role-based access control, or Azure RBAC. You can manage who has access to Azure resources by assigning roles.



[Source: Microsoft Documentation]

Create and assign custom roles [Azure roles and Microsoft Entra roles]

- **Azure Roles:** Azure uses role-based access control (RBAC) to manage access to resources. Roles define permissions, and assigning those roles to users or groups grants them the associated permissions.
  Microsoft Entra Roles: Microsoft Entra roles allow you to grant granular permissions to your administrators, adhering to the principle of least privilege. Microsoft Entra's built-in and custom roles operate on concepts similar to those you find in the role-based access control system for Azure resources (Azure Roles).

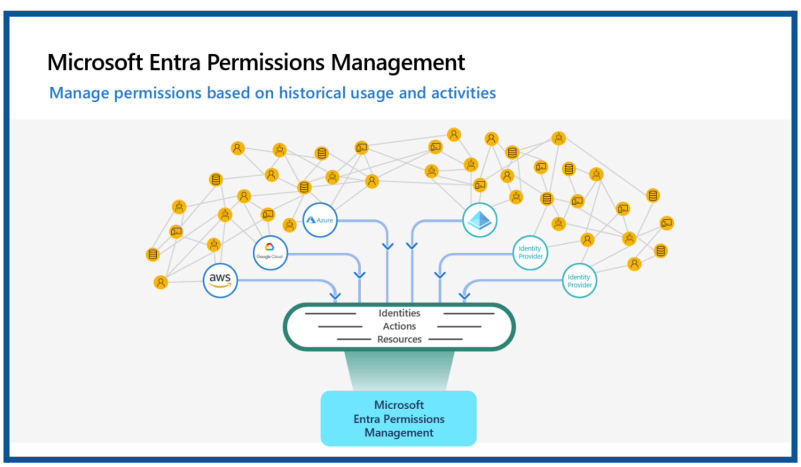
- **Built-in Azure Roles:** Owner, Contributor, and Reader.
  Owner: Full access to all resources, including the right to delegate access to others.
  Contributor: Can create and manage resources, but cannot grant permissions to others.
  Reader: View-only access to resources.

- **Custom Azure Roles:** Organizations often have specific needs not covered by built-in roles. Custom roles allow the definition of fine-grained permissions tailored to specific tasks. Components of a custom role include actions, notations, and assignableScopes.

  Source Links:
  [Create custom roles in Microsoft Entra role-based access control]
  [Create or update Azure custom roles using the Azure portal - Azure RBAC]

Implement and manage Microsoft Entra Permissions Management

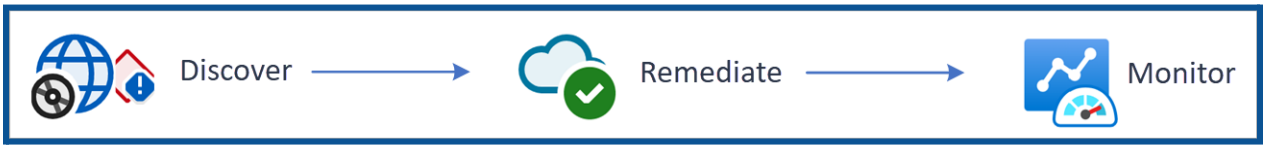
Microsoft Entra Permissions Management is a crucial component in ensuring secure and controlled access to resources within the cloud ecosystem.



[Source: Microsoft Documentation]

With Microsoft Entra Permissions Management, your organization can adopt a new, dynamic approach to managing identities and permissions in multi-cloud environments.

Permissions Management allows you to address three key use cases in securing and managing identity permissions in cloud environments: discover, remediate, and monitor.



[Source: Microsoft Documentation]

Configure Microsoft Entra Privileged Identity Management

Privileged Identity Management:

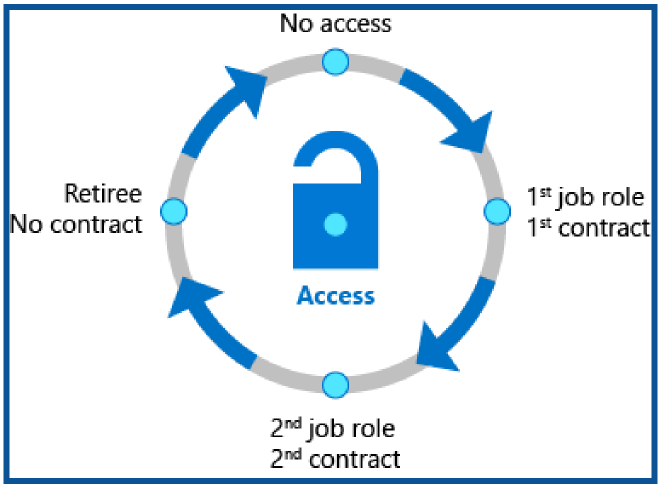
Privileged Identity Management (PIM) is a service in Microsoft Entra ID that enables you to manage, control, and monitor access to critical resources in your organization. These resources include resources from Microsoft Entra ID, Azure, and other Microsoft online services such as Microsoft 365 or Microsoft Intune.

- **Features of Privileged Identity Management**
  - Provides timely privileged access to Microsoft Entra ID and Azure resources
  - Specify timed access to resources using start and end dates
  - Approval is required to activate privileged roles
  - Enforce multi-factor authentication to activate any role
  - Use rationale to understand why users are active
  - Receive notifications when privileged roles are activated
  - Conduct access reviews to make sure users still need the roles
  - Download audit history for internal or external audits

Configure role management and access reviews

What are access reviews?

Access reviews in Microsoft Antra ID, part of Microsoft Antra, enable organizations to efficiently manage group membership, access to enterprise applications, and role assignments. User access can be regularly reviewed to ensure that only the right people continue to have access.



Source: [***Microsoft Entra ID Governance***]

The key benefits of enabling access reviews are:

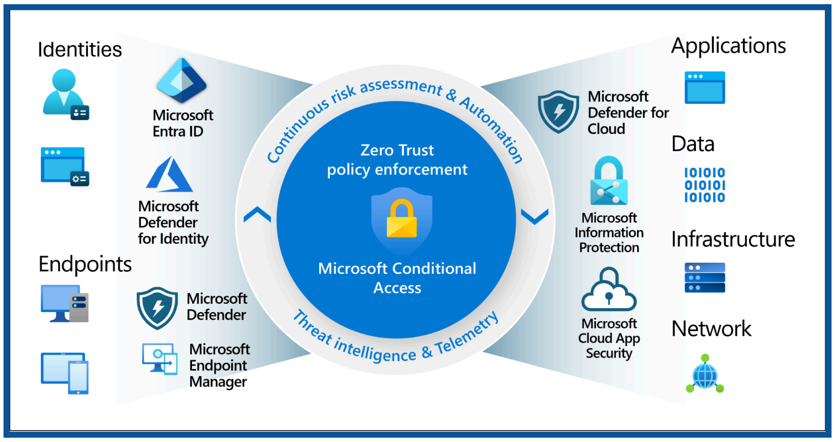
1. **Control collaboration:** Access reviews allow you to manage access to all the resources your users need. When users share and collaborate, you can be assured that information is only shared between authorized users.
2. **Manage risk:** Access reviews provide you with a way to review access to data and applications, reducing the risk of data leakage and spills. You gain the ability to regularly review external partners' access to corporate resources.
3. **Address compliance and governance:** With access reviews, you can control and recertify the access lifecycle for groups, apps, and sites. You can manage and track reviews for compliance or risk-sensitive applications unique to your organization.
4. **Reduce costs:** Access reviews are built in the cloud and work seamlessly with cloud resources like groups, applications, and access packages. Using access reviews is less costly than building your tools or otherwise upgrading your on-premises tool set.

Implement Conditional Access Policies

What is Conditional Access?

- Conditional Access is a security technique that limits access to resources or information based on specific conditions or criteria.
- Rather than giving unrestricted access to everyone, this method enables organizations to establish certain rules or standards that individuals must follow before being admitted.
- These conditions may include user roles, device compliance, location, or access time.

- Conditional access improves security by adjusting permissions based on established conditions, ensuring that only authorized users can access critical data or systems under specified circumstances.



Source: [***What is Conditional Access in Microsoft Entra ID?***]()

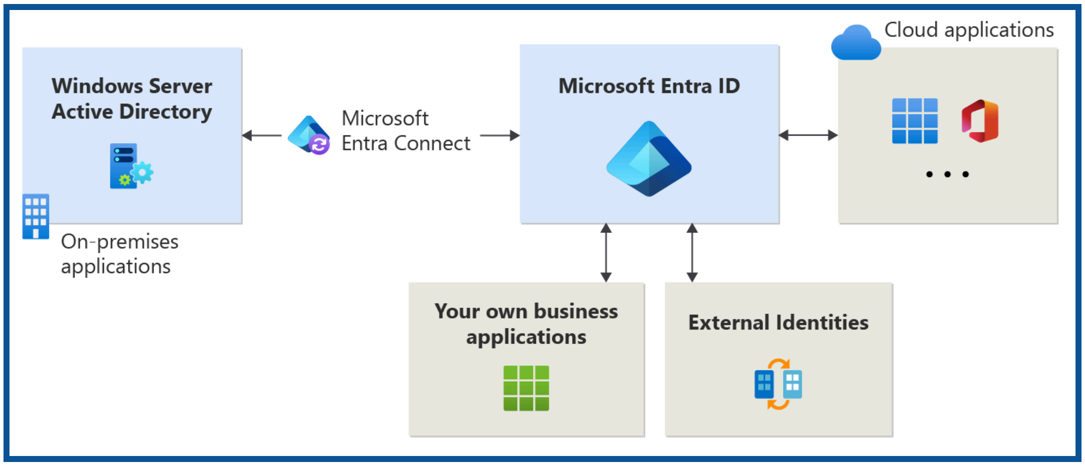
❖ Conditional Access is found in the Microsoft Entra admin center under **Protection > Conditional Access.**

D) Microsoft Entra Application Access

Manage access to enterprise applications in Microsoft Entra ID, includes OAuth permission grants

What is Application Management?

- Application management in Microsoft Entra ID refers to the creation, configuration, management, and monitoring of cloud-based applications.
- Authorized users can access an application securely if it is registered with a Microsoft Entra tenant.
- Many different types of applications can be registered in Microsoft Entra ID.

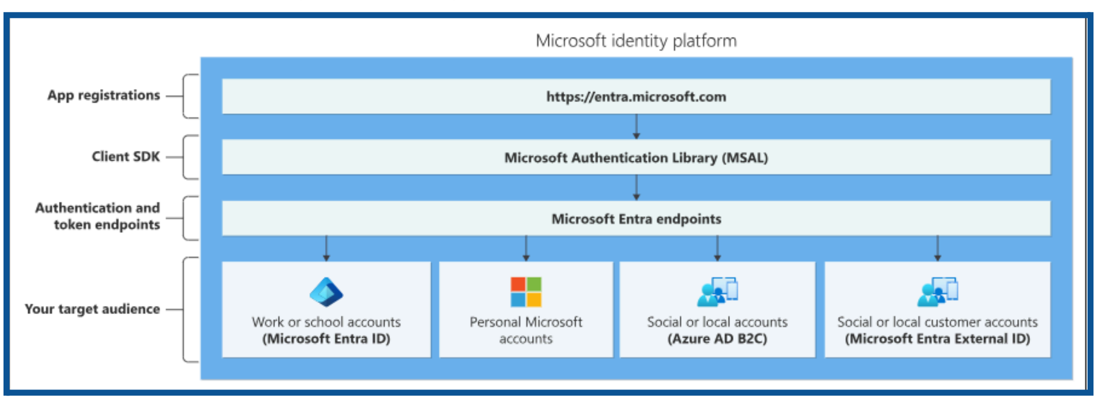


Source: [*What is application management? - Microsoft Entra ID*]

- Applications configured for federated single sign-on (SSO) with SAML-based authentication
- Application proxy applications that use Microsoft Entra Pre-Authentication
- Applications that are built on the Microsoft Entra Application Platform use OAuth 2.0/OpenID Connect authentication after a user or administrator has provided consent for that application.
- Some enterprise applications offer more control over who is allowed to sign in.

Manage Microsoft Entra app registrations

- When you register a new app with Microsoft Entra ID, a service principal is automatically created for the app registration.
- The service principal is the identity of the app in the Microsoft Entra tenant. Access to resources is restricted by roles assigned to the service principal, giving you control over which resources can be accessed and at what level.
- For security reasons, it is always recommended to use service principals with automated tools rather than allowing them to sign in with user credentials.



[Source: Microsoft Documentation]

Configure app registration permission scopes

- By registering your Web API and exposing it through scopes, assigning an owner and app role, you can provide permissions-based access to its resources to authorized users and client apps accessing your API.
- Roles and access scopes must be configured to access APIs. Set up roles and access scopes for your resource application web APIs if you wish to make them available to client applications. Set up permissions for a client application to access the web API during app registration if you want it to do so.
- The Microsoft identity platform must first register your web API before you can grant scoped access to its resources.

Manage app registration permission consent

- Managing app registration permission consent involves overseeing the process by which users grant or deny permission for an application to access their data or perform certain actions.
- This includes designing user interfaces that present permission requests, ensuring compliance with privacy regulations, such as GDPR or CCPA, tracking and storing consent preferences, and providing users with options to modify or revoke their consent settings as needed.
- Effective management of app registration permission consent is crucial for maintaining user trust and adhering to legal requirements regarding data privacy and security.

Types of Permissions

1. **Delegated permissions** are used in the Delegated Access scenario. They are permissions that allow applications to act on behalf of the user.
   The application will never be able to access anything that the signed-in user could not access himself. For example, imagine an application that has been granted the Files.Read.All delegated permission on behalf of the user, Tom.
   The application will only be able to read files that Tom can personally access

2. **Application permissions**, sometimes called app roles, are used in the apponly scenario, without a signed-in user present. The application will be able to access any data that the permission is associated with. For example, an application granted the Files.Read.All application permissions will allow you to read any file in the tenant. Only an administrator or owner of the service principal can consent to application permissions.

Manage and use service principals

If you register an application, an Application object and a Service Principal object are automatically created in your home tenant. If you register/create an application using the Microsoft Graph API, creating a service principal object is a separate step.

Service principal object

- To access resources protected by a Microsoft Entra tenant, the entity that requires access must be represented by a security principal.
- This requirement applies to users (user principals) and applications (service principals).
- The security principal defines the access policy and permissions for the user or application in the Microsoft Entra tenant.
- It enables core features such as authentication of the user or application during sign-in and authorization during resource access.

There are three types of service principles:

1. Application
2. Managed identity
3. Legacy

Manage managed identities for Azure resources

- Managed Identity Microsoft Entra automatically assigns a managed identity, the Microsoft Entra ID, to applications to use when connecting to resources that support authentication.
- Applications can use managed identity to obtain a Microsoft Entra token without managing any credentials.

There are two types of managed identities:

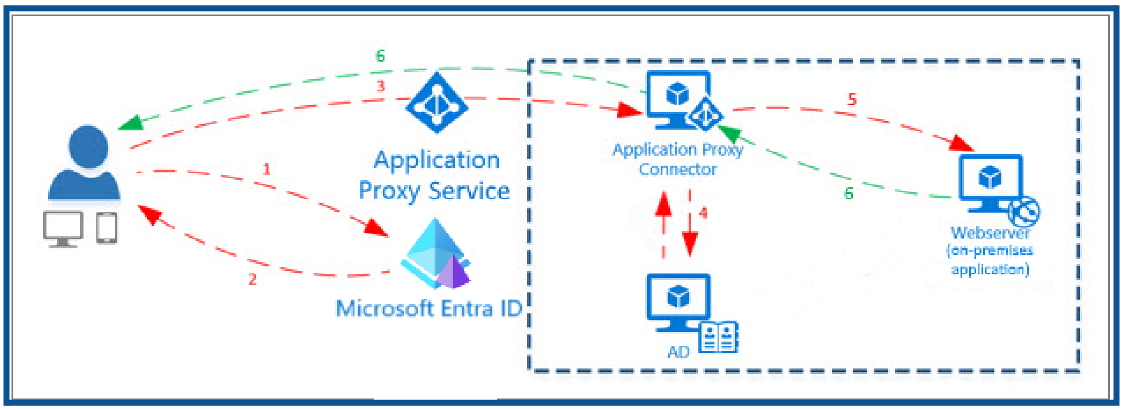
1. System-assigned
2. User-assigned

Benefits of using Managed Identities

- Keeping track of credentials is not required. Credentials are inaccessible to you.
- Managed identities can authenticate against any resource that supports Microsoft Entra authentication, including your applications.
- Managed identities are free to use.

What is an Application Proxy?

- Application Proxy is a feature of Microsoft Entra ID that enables users to access on-premises web applications from remote clients.
- Application proxying includes both application proxy services running in the cloud and application proxy connectors running on on-premises servers.
- Microsoft Entra ID, the Application Proxy service, and the Application Proxy Connector work together to securely pass the user sign-on token from the Microsoft Entra ID to the web application.



Source: *[Remote access to on-premises apps - Microsoft Entra application proxy]*

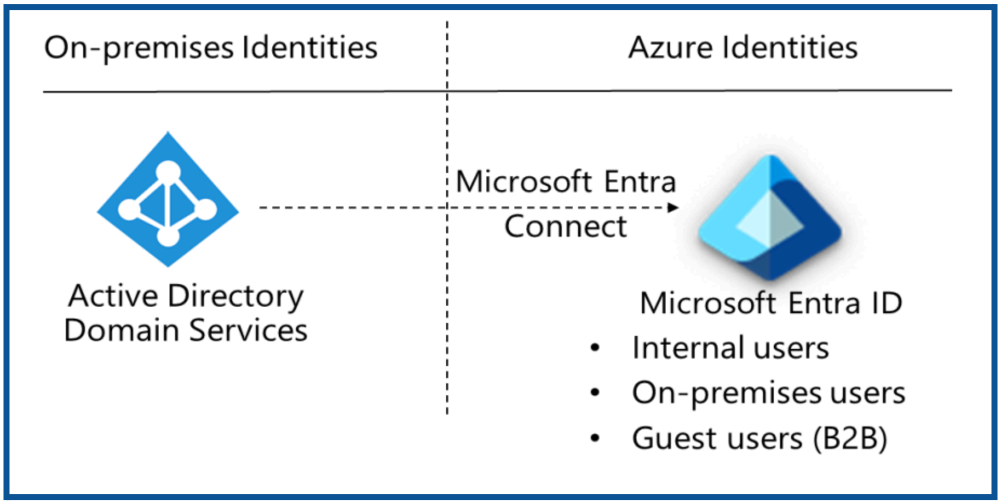
Microsoft Entra Application Proxy provides secure remote access to on-premises web applications. Once signed-on to a Microsoft Entra ID, users can access cloud and on-premises applications through an external URL or an internal application portal. For example, Application Proxy provides remote access and single sign-on to Remote Desktop, SharePoint, Teams, Tableau, Click, and line of business (LOB) applications.

Some key features of Microsoft Entra application proxy are as follows:

- It's Simple to use.
- It's very Secure.
- It's Cost-effective

For example, on-premises applications can use Conditional Access and two-step verification. Application proxy doesn't require you to open inbound connections through your firewall.

Microsoft Entra ID, Entra B2B, and Azure AD B2C



[Source: Microsoft Documentation]

What is Entra B2B?

Entra B2B stands for business-to-business. It's all about securely enabling collaboration and resource sharing with validated external entities (like partner organizations) while maintaining separate directories.

Use Case:

- Ideal for scenarios where your organization collaborates with other businesses.
- It remains unaffected by the introduction of Microsoft Entra External ID.
- Facilitates seamless interactions between different organizations.

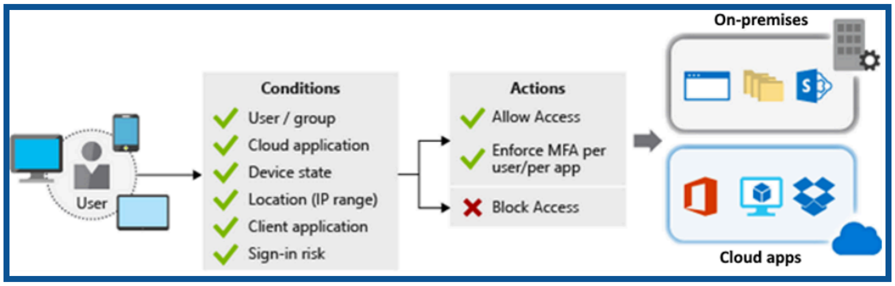
What is Azure AD B2C?

- Azure AD B2C is business-to-consumer. It focuses on managing identities for customer-facing applications.
- Perfect for apps that interact directly with end-users (like online shopping, social media, or travel booking).
- Create a separate Azure AD B2C tenant from your existing employee-based Microsoft Entra tenant.
- Azure AD B2C continues to be fully supported, and you can choose based on your immediate needs.

Microsoft Entra Conditional Access and Identity Protection

What is Microsoft Entra Conditional Access?

Conditional access is like a security bouncer for your digital club. It ensures that the right people (users) get into the right places (resources) at the right times, all while maintaining security.



[Source: Microsoft Documentation]

How does it work?

- **Signals:** It considers various signals, like user identity, device information, and risk factors.
- **Policies:** These are like "if-then" rules. For example, "If a user wants to access Microsoft 365, they must do multifactor authentication."
- **Decisions:** It can block or grant access based on conditions.
- To empower users while protecting your organization's assets.

In the Microsoft Entra admin center, under **Protection > Conditional Access.**

Access reviews managed identities,& service principals for applications

Access Reviews:

Access reviews are periodic evaluations of user access to resources (such as applications, files, or groups) within an organization. These reviews help ensure that access permissions remain appropriate over time.

- An administrator initiates an access review for a specific resource.
- Reviewers are assigned to assess the access rights of users.
- Reviewers evaluate whether each user's access is still necessary and appropriate.
- Based on the review, access can be approved, modified, or revoked.

Benefits:

- Regular reviews prevent unauthorized access.
- Aligns with regulatory requirements and Streamlines access management.

- Access reviews are commonly used for group memberships, application access, and role assignments.

Managed Identities:

Managed identities are automatically created within Azure resources (such as virtual machines, Azure Functions, or App Services) to authenticate with other Azure services.

Types:

- **System-assigned managed identity:** tied to a specific Azure resource and automatically deleted when the resource is removed.
- **User-assigned managed identity:** created independently and can be associated with multiple resources.

Benefits:

- **Simplified authentication:** no need to manage credentials.
- **Secure access:** Managed identities use Azure AD for authentication.
- **Seamless integration:** works seamlessly with Azure services.
- Managed identities are commonly used for accessing Azure Key Vault, Azure Storage, and Azure SQL Database.

Service Principals

Service principals are identities used by applications, services, or automation tools to authenticate and access resources.

Characteristics:

- **Non-human:** Unlike user accounts, service principals are not tied to individual users.
- **Credentials:** Service principals have client IDs and secret or certificate-based credentials.
- **Permissions:** Assign specific roles or permissions within Azure resources.

Use Cases:

- ➢ **Application Authentication:** Service principals allow applications to authenticate without user interaction.
- ➢ **Automated Processes:** Used for tasks like deploying resources or managing Azure subscriptions programmatically.
- ➢ **Access Control:** Service principals can be granted specific permissions to access resources.

Azure Key Vault

What is an Azure Key Vault?

Azure Key Vault is a cloud service designed to securely store and manage sensitive information, including:

- **Secrets:** such as API keys, passwords, and connection strings.
- **Cryptographic Keys:** are used for encryption, decryption, and signing.
- **Certificates:** for secure communication and authentication.

It eliminates the need to hardcode secrets and keys directly into application code, enhancing security and manageability.

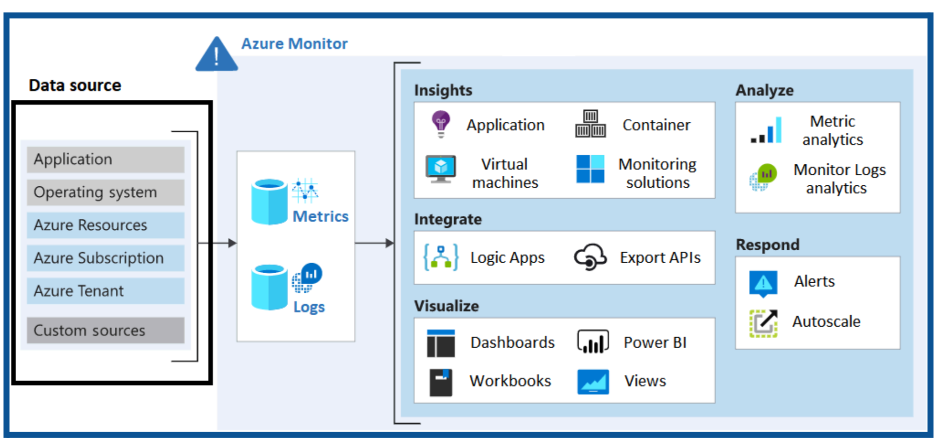
Containers:

- **Vaults:** These containers support storing both software and HSM-backed keys, secrets, and certificates.
- **Managed HSM Pools:** These containers exclusively support HSM-backed keys.

Design governance

What is Governance?

Azure Governance is a comprehensive framework that provides specialized guidelines for securing and managing cloud resources within Microsoft Azure.
It ensures security, compliance, cost optimization, and efficient management of Azure subscriptions and resources.



[Source: Microsoft Documentation]

Key Components:

- **Policies:** Define acceptable and unacceptable cloud activities.
- **Azure Blueprints:** Create fully governed environments with predefined templates.
- **Resource Graph:** Gain visibility into all your resources.
- **Cost Management:** Analyze costs and monitor usage.
- **Management Groups:** Organize and apply policies across multiple subscriptions.

Why is Azure governance important?

- **Security:** Establish guardrails to prevent unauthorized actions.
- **Compliance:** align with internal and external regulations.
- **Efficiency:** Optimize costs and resource utilization.
- **Auditing:** Maintain an audit trail for accountability.

Azure Governance Services:

Azure Policy: Enforce policies for any Azure service.
Azure Blueprints: Create compliant environments using templates.
Azure Resource Graph: Quickly explore all your resources.
Microsoft Cost Management: Optimize spend and analyze usage.

Azure Subscriptions, Management groups, Resource Groups and Tags

Azure Subscriptions:

An Azure subscription is the fundamental billing and access unit in Azure. It provides access to Azure services and resources.

- **Resource Isolation:** Each subscription acts as an isolated environment.
- **Billing Boundary:** All resources within a subscription are billed together.

Use Cases:

- **Project Isolation:** Different projects or teams can have separate subscriptions.
- **Billing Segmentation:** Separate billing for different departments or applications.

1. Resource Groups:

A resource group is a logical container for organizing and managing Azure resources.
Logical Grouping: Resources related to an application or project are grouped.
Resource Lifecycle: Resource groups allow you to manage resources collectively (create, update, delete).
Use Cases:

- **Application Components:** Group VMs, databases, storage accounts, etc., for an app.
- **Environment Segmentation:** Dev, test, and production environments.

2. Management Groups:

Management groups provide a hierarchical structure for managing access, policies, and compliance across multiple subscriptions.

Policy Enforcement: Apply policies consistently across subscriptions.
Access Control: Manage permissions at a higher level.

3. Tags:

➔ Tags are key-value pairs that you apply to Azure resources, resource groups, and subscriptions.

- **Metadata:** Helps identify resources based on relevant settings (e.g., environment, owner).
- **Cost Management:** Group costs for tagged resources.

Azure RBAC

Azure Role-Based Access Control (Azure RBAC) is a powerful authorization system built on Azure Resource Manager. It provides fine-grained access management for Azure resources.

- **Granular Access Control:** Manage who has access to Azure resources.
- **Resource Permissions:** Define what users can do with those resources.
- **Scoped Access:** Control access at different levels (management group, subscription, resource group, or resource).

Key Concepts:

1. **Security Principal:**
    a. Represents a user, group, service principal, or managed identity requesting access.
    b. Assign roles to these security principals.
2. **Role Definition:**
    a. A collection of permissions (actions like read, write, and delete).
    b. Built-in roles (e.g., owner, virtual machine reader) or custom roles.
3. **Scope:**
    a. The set of resources where access applies.

b. Specify at management group, subscription, resource group, or resource level.

→ **Use Cases:**
- Allow a user to manage virtual machines with a subscription.
- Permit a DBA group to manage SQL databases.
- Enable an application to access all resources in a resource group.

Azure Policy and Landing Zones

Azure Policy:

Azure Policy is a service that allows you to enforce rules and guidelines across your Azure resources.

- **Governance:** Ensure compliance with organizational standards and best practices.
- **Consistency:** Apply configurations consistently across resources.

Key Concepts:

1. **Policy Definitions:** Rules that specify what's allowed or denied.
2. **Assignments:** Application of policy definitions to specific scopes (subscriptions, resource groups, etc.).
3. **Built-in and Custom Policies:** Use predefined policies or create custom ones.

Use Cases:

- **Tagging Enforcement:** Require specific tags on resources.
- **Resource Locks:** Prevent accidental deletion of critical resources.
- **Allowed VM SKUs:** Limit VM sizes based on organizational guidelines.

Azure Landing Zones:

An Azure Landing Zone is an opinionated environment following key design principles across eight areas.

- **Scalable Environment:** accommodate application migration, modernization, and innovation at scale.
- **Resource Isolation:** Use subscriptions for application and platform resources.

Platform vs. Application Landing Zones:

Platform Landing Zone: Provides shared services (identity, connectivity, and management) to application landing zones.

Application Landing Zone: Contains application-specific resources.

Design data storage solutions.

Design data storage solutions for relational data

Azure SQL Database:

Azure SQL Database is a fully managed, cloud-based relational database service provided by Microsoft Azure.

Features:

- **Database Compatibility:** It supports compatibility levels ranging from 100 to 160.
- **Always Encrypted:** Yes, it provides support for Always Encrypted using certificate stores and key vaults.
- **Always On Availability Groups:** It guarantees 99.99% to 99.995% availability for every database. Disaster recovery options are discussed in the Overview of Business Continuity with Azure SQL Database.
- **Active Geo-Replication:** Available across all service tiers.
- **Auto-Scale:** Available in the serverless model; in the non-serverless model, service tier changes (such as vCore, storage, or DTU) can be done quickly and online.

Azure SQL Managed Instance:

Azure SQL Managed Instance is a fully managed SQL Server Database Engine instance hosted in Azure and placed within your network.

Features:

- **Compatibility:** It offers nearly 100% compatibility with on-premises SQL Server databases.
- **Always Encrypted:** Supports Always encrypt using certificate stores and key vaults.
- **Always On Availability Groups:** This guarantees 99.99% availability for every database and cannot be directly managed by users. Disaster recovery details are covered in the Overview of Business Continuity with the Azure SQL Database. Failover groups can be used to configure a secondary SQL Managed Instance in another region.
- **Active Geo-Replication:** Not available; failover groups serve as an alternative.
- **Auto-Scale:** Split compute and storage components.

SQL Server on Azure Virtual Machines:

SQL Server on Azure Virtual Machines provides complete control and customization for organizations needing to support legacy applications or specific infrastructure requirements.

Features:

- **Deployment Model:** It allows you to lift and shift on-premises applications to the cloud with minimal application and database changes.
- **Compute and Storage:** You have full control over compute and storage components.
- **High Availability:** While not inherently managed by Azure, you can configure high availability using traditional SQL Server features.
- **Customization:** Ideal for scenarios where legacy applications require specific infrastructure configurations

Azure SQL Edge and Azure Cosmos DB

Azure SQL Edge:

Azure SQL Edge is a robust Internet of Things (IoT) database designed for edge computing.
It combines several capabilities:

- **Data Streaming:** Azure SQL Edge supports data streaming, allowing real-time processing of data from IoT devices.
- **Time Series Data:** It handles time series data efficiently.
- **Built-in Machine Learning:** Azure SQL Edge integrates machine learning features directly within the database engine.
- **Graph Features:** It also provides graph capabilities.

Deployment Flexibility:

- You can deploy Azure SQL Edge across various environments:
  - **Edge Devices:** Extend the Microsoft SQL engine to edge devices for consistent performance and security.
  - **On-Premises Datacenters:** Deploy applications once and run them anywhere.
  - **Azure Cloud:** Seamlessly integrate with your cloud-based solutions.

Key Features:

- **Low-Latency Analytics:** Process data at the edge to overcome latency constraints.
- **Security:** Enjoy turnkey management and security features.
- **Scalability:** simplified pricing with no upfront costs.

Use Cases: Ideal for scenarios involving IoT data, real-time analytics, and edge computing.

Azure Cosmos DB:

Azure Cosmos DB is a globally distributed, multi-model database-as-a-service.

- **Key features include:**
  - **NoSQL Foundation:** It provides NoSQL capabilities with comprehensive SLAs on throughput, availability, consistency, and latency.
  - **Multi-Model Support:** Exposes data through SQL, MongoDB, Cassandra, Table, and Gremlin APIs.
  - **Global Distribution:** Manage data across scattered data centers worldwide.

Core Attributes:

- **Low Latency:** single-digit millisecond response times.
- **Scalability:** automatic and instant scalability.
- **Guaranteed Speed:** Ensures performance at any scale.
- **High Availability:** SLA-backed availability.
- **Enterprise-Grade Security:** Transparent data encryption, data masking, and Always Encrypted.

Use Cases: Widely used in serverless applications requiring low-latency responses and rapid global scaling.

Database Scalability and Availability

1. Scalability:

Scalability refers to the ability of a system to handle an increased workload or growing demands without compromising performance. In the context of databases, scalability can be achieved through various mechanisms:

a. Vertical Scalability (Scaling Up):

- Vertical scalability involves adding more resources (CPU, memory, storage) to a single database server.

- Common approaches include upgrading hardware, increasing memory, or adding faster storage.
- While it's straightforward, there are limits to how much a single server can scale vertically.
- Use cases: Small to medium-sized workloads with predictable growth.

b. Horizontal Scalability (Scaling Out):

- Horizontal scalability involves distributing the workload across multiple database servers.
- Common approaches include sharding (partitioning data across servers) and using database clusters.
- Benefits: Improved fault tolerance, better performance, and the ability to handle large-scale workloads.
- Use cases: High-traffic applications, big data scenarios, and globally distributed systems.

2. Availability:

Availability ensures that a database remains accessible and operational even during failures or maintenance. Achieving high availability involves several strategies:

a. Replication:

- Replication involves maintaining multiple copies (replicas) of the database.
- Types of replication:
    - **Master-Slave Replication**: One master for writes, multiple read-only slaves.
    - **Multi-Master Replication**: Multiple masters for both reads and writes.
- Benefits: Improved read performance, failover capabilities, and load balancing.
- Use cases: Read-heavy workloads, disaster recovery.

b. Failover and High Availability Groups:

- Implement failover mechanisms to automatically switch to a standby server in case of primary server failure.
- High Availability Groups (HAGs) provide automatic failover and load balancing.
- Use cases: Critical applications, minimizing downtime.

c. Geo-Replication:

- Geo-replication involves maintaining copies of the database in different geographic regions.
- Provides disaster recovery capabilities and improves performance for globally distributed users.
- Use cases: Global applications, compliance requirements.

3. Considerations:

- **SLAs (Service Level Agreements)**: Understand the availability requirements and design accordingly.
- **Data Consistency**: Choose the right replication method based on consistency needs (strong vs. eventual consistency).
- **Cost vs. Benefit**: Evaluate the trade-offs between scalability, availability, and cost.

Data Encryption for Structured Data

Data Encryption for Structured Data

1. Encryption at Rest:

- **Azure Storage Service Encryption (SSE)** ensures that all data stored in Azure is encrypted.
- Key points:
  - **256-bit Advanced Encryption Standard (AES)** cipher is used.
  - Compliant with **FIPS 140-2** security standards.
  - It applies to structured data stored in Azure storage accounts, databases, and other services.
  - Protects against unauthorized access and data breaches.

2. Types of Encryption:

- Symmetric Encryption:
  - Uses a single secret key for both encryption and decryption.
  - Efficient for large-scale data.
  - Examples: AES, DES.
  - Suitable for structured data protection.
- Asymmetric Encryption:
  - Involves a pair of keys: public and private.
  - The public key encrypts the private key decrypts.
  - Ensures secure communication and data exchange.
  - Examples: RSA, ECC.
  - Useful for scenarios like a secure key exchange.

3. Best Practices:

- **Key Management:**
  - Properly manage encryption keys.
  - Use Azure Key Vault for centralized key storage and management.
  - Rotate keys periodically.
- **Transport Encryption:**
  - Encrypt data in transit using **TLS/SSL** protocols.
  - Applies to structured data during communication between clients and services.
- **Data Masking:**
  - Mask sensitive data within structured data.
  - Useful for compliance and privacy requirements.
  - Examples: SSN, and credit card numbers.
- **Database-Level Encryption:**
  - Implement encryption at the database level.
  - Transparent Data Encryption (TDE) for SQL databases.
  - Always encrypt sensitive columns.

Design data storage solutions for semi-structured and unstructured data

Azure Data Storage Services and Storage Accounts

Azure provides a range of storage services to meet diverse data storage requirements. Understanding these services is crucial for designing effective solutions:

- **Azure Blob Storage:**
  - Blob storage is ideal for storing unstructured data such as images, videos, and documents.
  - Key features:
    - **Block Blobs**: Suitable for large files.
    - **Page Blobs**: Used for virtual machine disks.
    - **Append Blobs**: Designed for append-only scenarios (e.g., log files).
- **Azure Table Storage:**
  - A NoSQL data store for semi-structured data.
  - Stores data in tables with a schema-less design.
  - Useful for scenarios like logging and sensor data.
- **Azure Queue Storage:**
  - Provides a message queue for asynchronous communication between components.

- Commonly used for decoupling workloads and ensuring reliable message processing.

- **Azure File Storage:**
  - Offers fully managed file shares accessible via the SMB protocol.
  - It is useful for migrating on-premises applications to the cloud.

Storage Accounts

1. **Deployment Models:**
   - Azure storage accounts can be created in two deployment models:
     - **General Purpose v2 (GPv2)**: Supports all storage services (blobs, tables, queues, and files).
     - **Blob Storage**: Optimized for blob storage scenarios.
2. **Redundancy Options:**
   - Choose from the following redundancy options:
     - **Locally redundant storage (LRS)**: data replicated within the same data center.
     - **Geo-Redundant Storage (GRS)**: data replicated to a secondary region for disaster recovery.
     - **Zone-Redundant Storage (ZRS)**: data replicated across availability zones within a region.
3. **Storage Account Options:**
   - **Standard**: Suitable for most workloads.
   - **Premium** high-performance storage for virtual machine disks.
   - **Cool:** cost-effective storage for infrequently accessed data.
   - **Archive** low-cost storage for long-term retention.
4. **Storage Types:**
   - **Blob Storage**: For unstructured data.
   - **File Storage**: SMB-based file shares.
   - **Queue Storage**: MessageQueuess.
   - **Table Storage**: Semi-structured NoSQL data.
5. **Moving Files:**
   - Use tools like **AzCopy**, **Azure Storage Explorer**, or **Azure File Sync** to move files between on-premises and Azure storage.

Azure Blob Storage

Azure Blob Storage is a versatile service for storing unstructured data, such as images, videos, backups, and logs. Here are the key points you need to know:

1. **Blob Types:**
    - **Block Blobs: Ideal for large files (up to 4.75 TB). Commonly used for media files, backups, and logs.**
    - **Page Blobs: Used for virtual machine disks (VHDs). Provides random read/write access.**
    - **Append Blobs: Designed for scenarios where data is appended sequentially (e.g., log files).**

2. **Access Tiers:**
    - Blob storage offers different access tiers based on data access patterns:
    - **Hot**: frequent access with low latency. Suitable for active data.
    - **Cool**: infrequent access with lower storage costs. Ideal for backups and archives.
    - **Archive**: lowest-cost tier for long-term retention. Data retrieval takes longer.

3. **Security and encryption:**
    - **Encryption at Rest**:
    - All data stored in Azure Blob Storage is automatically encrypted.
    - Uses **256-bit Advanced Encryption Standard (AES)**.Compliant with **FIPS 140-2** security standards.

4. **Shared Access Signatures (SAS):**
    - Generate SAS tokens to grant time-limited access to specific blobs or containers.
    - Fine-grained control over permissions (read, write, delete, etc.).

5. **Lifecycle Management:**
    - Define rules to automatically transition blobs between access tiers.
    - For example, move data from the “hot” tier to the “cool” tier after a certain period.

6. **Blob Indexing:**
    - Enables efficient querying of blob metadata and properties.
    - Use Azure Search or Azure Cognitive Search for advanced indexing and searching.

7. **Versioning:**
    - Enable versioning to maintain historical versions of blobs.
    - Useful for compliance and auditing purposes.

8. **Monitoring and Metrics:**
    - Monitor blob storage using Azure Monitor.
    - Metrics include ingress/egress data, availability, and latency.

Azure Storage - Data redundancy

Azure Storage always ensures that your data is protected from both planned and unplanned events. Redundancy plays a crucial role in meeting availability and durability targets, even in the face of failures.

1. Redundancy in the Primary Region

In the primary region, Azure Storage replicates your data to ensure its safety:

- ➔ Locally Redundant Storage (LRS):
  - ◆ LRS synchronously replicates your data three times within a single data center in the primary region.
  - ◆ While it's the least expensive option, it's not recommended for applications requiring high availability or durability.
- ➔ Zone-Redundant Storage (ZRS):
  - ◆ ZRS synchronously copies your data across three Azure availability zones within the primary region.
  - ◆ Recommended for applications that demand high availability.

2. Redundancy Across Regions (Geo-Replication)

To protect against regional disasters, consider replicating your data to a secondary region:

- ➔ Geo-Redundant Storage (GRS):
  - ◆ GRS asynchronously replicates your data to a secondary region, which is geographically distant from the primary region.
  - ◆ Provides additional protection against regional outages.
  - ◆ Read access to the replicated data is not available in the secondary region.
- ➔ Geo-Redundant Storage with Read Access (RA-GRS):
  - ◆ Similar to GRS but allows read access to the replicated data in the secondary region.
  - ◆ Useful for scenarios where continuity is critical.

3. Recommendations

- Choose ZRS in the Primary Region:
  - For applications requiring high availability, use ZRS within the primary region.
  - Consider this option for Azure Data Lake Storage Gen2 workloads.
- Combine GRS or RA-GRS with ZRS:
  - For maximum protection, replicate your data to a secondary region using GRS or RA-GRS.
  - This ensures continuity even if the primary region becomes unavailable.

Azure Files and Table Storage

Azure Files

Azure Files provides fully managed file shares accessible via the SMB (Server Message Block) protocol. Here are the key points you need to know:

- **Use Cases:**
  - Migrate on-premises applications to the cloud without significant code changes.
  - Share files across multiple virtual machines (VMs) or services.
  - Store configuration files, logs, and user data.

- **Features**:
  - **SMB Protocol**: accessible from Windows, Linux, and macOS using standard file system APIs.
  - **REST API**: Allows programmatic access.
  - **Encryption at Rest**: Data stored in Azure Files is encrypted.
  - **Access Control**: Use Azure Active Directory (Azure AD) for authentication and authorization.

- **Performance Tiers:**
  - **Standard**: Suitable for most workloads.
  - **Premium**: High-performance storage for VM disks.

Azure Table Storage

Azure Table Storage is a NoSQL data store for semi-structured data. Here's what you need to know:

- **Data Model:**
  - Stores data in tables with a schema-less design.
  - Each table contains entities (rows), and each entity has properties (columns).
  - Suitable for scenarios like logging, sensor data, and metadata storage.

- **Key Features:**
  - **Partition Key and Row Key**: Together, they uniquely identify an entity within a table.
  - **Scalability**:horizontally scalable to handle large amounts of data.
  - **Query Capabilities**: Supports basic queries based on partition and row keys.
  - **Secondary Indexes**: Create secondary indexes for efficient querying.

- **Use Cases:**
  - Storing large amounts of data with minimal schema requirements.
  - Building applications that require fast read and write access.

Azure Disk Solution and Storage Security

Azure Disk Solutions

1. Azure Managed Disks

- **Purpose**: Azure Managed Disks simplify disk management for virtual machines (VMs).
- **Features**:
  - Automatically replicated within a region for redundancy.
  - Supports both standard HDDs and premium SSDs.
  - Simplifies scaling and resizing VM disks.
  - Ideal for VM workloads requiring high availability and durability.

2. Disk Encryption

- **Azure Disk Encryption**:
  - Encrypts OS and data disks using Azure Key Vault.
  - Uses BitLocker for Windows VMs and DM-Crypt for Linux VMs.
  - Ensures data confidentiality and integrity.
  - Mandatory for compliance requirements.

3. Storage Security

a. Shared Access Signatures (SAS)

- Generate SAS tokens to grant time-limited access to specific resources (blobs, files, and queues).
- Fine-grained control over permissions (read, write, delete).
- It is useful for sharing data securely with external parties.

b. Azure Private Link

- Securely access Azure services over a private network.
- Routes traffic through a private IP address.
- Isolates services from the public internet.
- Enhances security and compliance.

c. Network Security Groups (NSGs)

- Control inbound and outbound traffic to Azure resources.
- Define rules based on source/destination IP, port, and protocol.
- Protects VMs, storage accounts, and other services.

d. Role-Based Access Control (RBAC)

- Assign granular permissions to users and groups.
- Control access to Azure resources.
- Follow the principle of least privilege.

e. Azure Firewall and Virtual Network Service Endpoints

- Azure Firewall: Provides network-level protection for Azure resources.
- Virtual Network Service Endpoints: Extend private IP addresses to Azure services.
- Enhance security by restricting access to specific VNets.

4. Considerations

- **Data Classification**: Classify data based on sensitivity (public, confidential, etc.).
- **Data Retention Policies**: Define retention periods and deletion rules.
- **Monitoring and Auditing**: Use Azure Monitor and Azure Security Center.

Design data integration

Azure Data Factory, Azure Data Lake, and Azure Databricks

Azure Data Factory (ADF)

Azure Data Factory is a cloud-based data integration service that allows you to create, schedule, and manage data-driven workflows. Here are some key points about ADF:

1. Data Movement and Orchestration:
    - ADF enables you to move data between various data stores (on-premises, cloud, and hybrid) using pipelines.
    - Pipelines consist of activities (e.g., copy data, transform data) orchestrated sequentially.
2. Integration with Azure Services:
    - ADF integrates seamlessly with other Azure services such as Azure SQL Database, Azure Blob Storage, and Azure Data Lake.
    - Use linked services to connect to these data sources.
3. Data Transformation:
    - ADF supports data transformation using data flows.
    - Data flows allow you to transform, clean, and enrich data within ADF.

Azure Data Lake

Azure Data Lake Storage Gen2 (ADLS Gen2) is a scalable and secure data lake solution. Here are key points about ADLS Gen2:

1. **Hierarchical File System:**
   - ADLS Gen2 provides a hierarchical file system on top of Azure Blob Storage.
   - Organize data into directories and files for efficient management.
2. **Scalability and Performance:**
   - ADLS Gen2 scales to petabytes of data.
   - Supports parallel access for big data workloads.
3. **Security and encryption:**
   - Data in ADLS Gen2 is encrypted at rest.
   - Use Azure AD for authentication and fine-grained access control.

Azure Databricks

Azure Databricks is an Apache Spark-based analytics platform. Here's what you need to know:

1. Unified Analytics Platform:
   - It combines data engineering, data science, and business analytics.
   - Provides notebooks for interactive data exploration and machine learning.
2. Delta Lake Integration:
   - Delta Lake is a storage layer that brings ACID transactions to data lakes.
   - Use Delta Lake with Azure Databricks for reliable data pipelines.
3. ETL and Data Engineering:
   - Azure Databricks notebooks allow you to build ETL (Extract, Transform, Load) pipelines.
   - Use PySpark or Scala for data transformations.
4. Machine Learning and AI:
   - Train machine learning models using Databricks notebooks.
   - Leverage MLflow for model tracking and deployment.

Azure Synapse Analytics & Data Analysis Solutions

Azure Synapse Analytics

Azure Synapse Analytics is a powerful analytics service that brings together enterprise data warehousing and big data analytics. Here are some key points about Synapse Analytics:

➔ **Unified Analytics Platform:**
   - ◆ It combines data engineering, data science, and business analytics.
   - ◆ Provides end-to-end solutions for data preparation, data management, data warehousing, and AI tasks.

- **Limitless Data Insights:**
  - Derive accurate, granular insights from your data using real-time streaming, data analysis, and machine learning techniques.
  - Keep your data safe using security and privacy features.

- **Components:**
  - **SQL Serverless**: On-demand query execution for ad-hoc analysis.
  - **Spark Pools**: Scalable big data processing using Apache Spark.
  - **Data Lake Storage Gen2 Integration**: Seamlessly analyze Azure Data Lake Storage Gen22 data.

Azure Stream Analytics

Azure Stream Analytics provides real-time data processing capabilities. Here's what you need to know:

1. Ingestion and Processing:
   - Ingest streaming event data from various sources (IoT devices, logs, social media) into Azure Synapse Analytics for further analysis and reporting.
   - Process data in real-time using SQL-like queries.
2. Common Scenarios:
   - Real-time dashboards: Visualize real-time data with Azure Stream Analytics and tools like Power BI.
   - Anomaly detection: identify patterns and anomalies in streaming data.
   - Event-driven alerts: trigger alerts based on specific conditions in the data stream.
3. Integration with Azure Synapse Analytics:
   - Azure Stream Analytics jobs can output to a dedicated SQL pool table in Azure Synapse Analytics.
   - Enables seamless data flow from real-time streams to analytical insights

Hot, Warm, and Cold Data Paths

In data management and analytics, the concept of hot, warm, and cold data paths refers to how data is handled based on its access patterns, latency requirements, and storage costs. Let's break down each path:

Hot Path:

- **Purpose**: for processing or displaying data in real-time.
- **Characteristics**:
  - Real-time alerting and streaming operations are performed using this data.
  - Requires very low latency.
- **Azure Services**:
  - Azure Functions: Provides a consumption-based and elastic resource to ingest incoming data for processing and alerting.
  - Azure SignalR: Enables real-time data streaming through WebSocket-based connections.
  - Azure App Service: Hosts web apps for displaying real-time data.

Warm Path:

- **Purpose**: For storing or displaying a recent subset of data.
- **Characteristics**:
  - Small analytic and batch-processing operations are performed on this data.
  - Typically covers the last 24 hours' worth of data.
- **Azure Services**:
  - Azure App Service: Allows querying and displaying recent data from Azure Data Explorer (formerly known as Kusto) or other storage solutions.

Cold Path:

- **Purpose**: For long-term storage of data.
- **Characteristics**:
  - Time-consuming analytics and batch processing are performed on this data.
  - Efficiently stores data for extended periods (e.g., default of 100 years).
  - Typically used for historical analysis and compliance.
- **Azure Services**:
  - Azure Data Explorer: efficiently stores and queries large volumes of data using the Kusto Query Language (KQL).

Design Business Continuity Solutions

Design solutions for backup and disaster recovery

Describe the recovery time objective and recovery point objective

Recovery Time Objective (RTO)

RTO refers to the Recovery Time Objective, which is the targeted duration of time and a service level within which a business process must be restored after a disaster or disruption to avoid unacceptable consequences associated with a break in business continuity.

Key Points:

- Time-focused: RTO is all about the time it takes to recover after an incident.
- Business Continuity: It ensures that critical operations resume within the defined time frame.
- Planning: This involves detailed planning for disaster recovery strategies.

Recovery Point Objective (RPO):

RPO stands for Recovery Point Objective, indicating the maximum tolerable period in which data might be lost from an IT service due to a major incident.

Key Points:

- Data-Focused: RPO deals with the amount of data that can be lost without significant harm to the business.
- Backup Frequency: Determines how often data backups should occur.
- Cost Implications: A lower RPO can mean higher costs due to more frequent backups.

RTO vs. RPO: The Differences

- While RTO and RPO may seem similar, they serve different purposes in a disaster recovery plan:
- RTO is concerned with the recovery of services and how quickly they need to be up and running post-disruption.
- RPO focuses on data loss and defines the acceptable age of files that must be recovered from backup storage for normal operations to resume.

Both RTO and RPO are crucial in developing a robust disaster recovery plan.

- ➔ They help organizations prepare for the worst and ensure minimal impact on operations.
- ➔ Balancing the two objectives can be challenging, as tighter objectives can lead to higher costs.
- ➔ Therefore, businesses need to assess their priorities and resources to set realistic and effective RTO and RPO targets.

Explore high availability and disaster recovery options

Exploring High Availability and Disaster Recovery Options

In the IT domain, ensuring systems' continuous operation and data protection against loss in the event of a disaster is paramount. This is where the concepts of high availability (HA) and disaster recovery (DR) come into play.

High Availability (HA):

High availability is the design approach to ensuring that a system or service is available for as long as possible, minimizing downtime, and ensuring that users have continuous access to the required IT resources.

Key characteristics of HA:

- **Redundancy:** HA systems are designed with redundant components to avoid single points of failure.
- **Failover Mechanisms:** Automatic failover processes ensure that if one component fails, another immediately takes over to maintain service continuity.
- **Load Balancing:** Distributing workloads across multiple systems to ensure no single server is overwhelmed and to provide seamless service to users.

Disaster Recovery (DR):

Disaster recovery, on the other hand, is the strategic plan and set of procedures put in place to recover IT systems, data, and infrastructure to operational status following a disaster.

Key Components of DR:

- **Data Backups:** Regular backups of data to secure locations ensure that data can be restored from a point before the disaster occurs.
- **Recovery Plans:** Detailed and tested plans that outline the steps to be taken in the event of a disaster to restore operations.
- **Site Recovery:** The use of alternate physical or cloud-based sites where operations can be transferred if the primary site is compromised.

Combining HA and DR for Business Continuity

For a comprehensive business continuity strategy, HA and DR are often combined. HA can handle most system failures by providing a reliable and resilient infrastructure, while DR is reserved for more catastrophic events that require a broader recovery effort.

Deployment architectures for HA and DR:

- **SQL Server on Azure VMs:** Offers a range of business continuity solutions, including Always On availability groups and failover cluster instances.
- **Multi-Server Clusters:** Advanced HA systems use multi-server clusters to provide redundancy and ensure service availability even if one cluster fails.
- **Azure Site Recovery:** A service that contributes to DR by orchestrating replication, failover, and recovery of virtual machines and physical servers.

High availability and disaster recovery are two sides of the same coin, both essential for maintaining service uptime and protecting data.

While HA focuses on preventing downtime, DR prepares for the worst-case scenario, ensuring that services can be restored after significant disruptions.

Implementing both HA and DR is crucial for any organization that relies on continuous system availability and data integrity.

Azure's HADR - VM Features

High Availability (HA) for Azure VMs

Availability Sets (ASs)

- **Availability sets** provide VM redundancy and availability within a data center by distributing VMs across multiple isolated hardware nodes.
- A subset of VMs keeps running during planned or unplanned downtime, ensuring the entire app remains available and operational.
- ASs are suitable for scenarios where you need basic HA within a single datacenter.

Availability Zones (AZs)

- **Availability zones** are unique physical locations that span data centers within an Azure region.
- Each AZ has an independent power, cooling, and networking infrastructure.
- Azure regions with AZs have a minimum of three separate AZs.

- Deploying VMs across AZs protects them from data center failures.
- For HA apps, it's recommended to use AZs whenever possible, as cross-zone HA provides >99.99% SLA due to resilience against data center failures.

Proximity Placement Groups (PPGs)

- If app latency is a primary concern, consider using **proximity placement groups** (PPGs) alongside AZs and ASs.
- PPGs allow you to colocate services in a single data center, improving communication latency between VMs.

Disaster Recovery (DR) Options

Azure Site Recovery (ASR)

- **Azure Site Recovery** is a comprehensive DR solution.
- It replicates VMs and physical servers from a primary site to a secondary site (often in a different Azure region).
- ASR provides regional DR and can also replicate data natively if the app supports it.
- Note that replication between regions is asynchronous, so some data loss may occur.

Multi-Region DR

- For more robust DR, consider multi-region setups.
- Use a global load balancer (e.g., Azure Front Door or Traffic Manager) to distribute traffic across multiple Azure regions.
- This approach ensures high availability even if an entire region becomes unavailable.

PaaS Deployments HADR options

Active-Active Strategy

An **active-active** strategy involves deploying your PaaS services across multiple Azure regions or data centers. Each region hosts an instance of your application, and traffic is distributed across these instances.

Benefits:

- **High availability**: Even if one region experiences an outage, the other regions continue serving traffic.
- **Load distribution**: Distributing user requests across regions optimizes resource utilization.
- **Reduced latency**: Users access the nearest region, minimizing latency.

Considerations:

- **Data synchronization**: Ensure data consistency across regions using techniques like geo-replication or event-driven data propagation.
- **Cost**: Running multiple instances can increase costs, so evaluate based on your workload.

Recommendations by Service Type

1. **Azure App Service**:
   - Use **App Service Environments (ASEs)** for HA.
   - ASEs provide a dedicated, isolated environment for running your apps.
   - Deploy ASEs across multiple regions for redundancy.
2. **Azure Functions**:
   - Leverage **Premium Plan** with **Regional VNET Integration**.
   - Deploy functions in multiple regions for HA.
3. **Azure Logic Apps**:
   - Use **Standard Logic Apps** with **Regional Deployment**.
   - Deploy logic apps across regions.
4. **Azure Cosmos DB**:
   - Enable **multi-region writes** to replicate data across regions.
   - Configure **failover priorities** to control which region becomes primary during failover.
5. **Azure SQL Database**:
   - Use **Geo-Replication** to replicate databases across regions.
   - Implement **read-scale replicas** for HA and offload read traffic.
6. **Azure Cache for Redis**:
   - Enable **replication** across regions.
   - Use **Azure Traffic Manager** for load balancing.
7. **Azure SignalR Service**:
   - Deploy SignalR Service in multiple regions.
   - Use **Azure Front Door** for global load balancing.

Disaster Recovery (DR) Options

1. **Azure Site Recovery (ASR)**:
   - ASR supports PaaS services.
   - Replicate PaaS resources to a secondary region.
   - Provides regional DR and asynchronous replication.
2. **Multi-Region DR**:
   - For robust DR, consider multi-region setups.
   - Use global load balancers to distribute traffic across regions.
   - Ensure data consistency and failover readiness.

Explore an IaaS high availability and disaster recovery solution

High Availability (HA) for IaaS Apps

Availability Sets (ASs)

- **Availability sets** (ASs) play a crucial role in achieving HA within a single data center.
- By distributing VMs across multiple isolated hardware nodes, ASs ensure VM redundancy and availability.
- During planned or unplanned downtime, a subset of VMs continues running, ensuring the entire app remains operational.
- ASs are suitable for scenarios where basic HA within a single datacenter is required.

Availability Zones (AZs)

- **Availability zones** (AZs) take HA to the next level by spanning unique physical locations within an Azure region.
- Each AZ has an independent power, cooling, and networking infrastructure.
- Azure regions with AZs consist of a minimum of three separate AZs.
- Deploying VMs across AZs protects them from data center failures.
- For HA apps, prioritize using AZs whenever possible. Cross-zone HA provides an impressive SLA of >99.99% due to resilience against data center failures.

Proximity Placement Groups (PPGs)

- If minimizing app latency is critical, consider using **proximity placement groups** (PPGs) alongside AZs and ASs.
- PPGs allow you to colocate services within a single data center, improving communication latency between VMs.

Disaster Recovery (DR) Options

Azure Site Recovery (ASR)

- **Azure Site Recovery** (ASR) is a comprehensive DR solution.
- It replicates VMs and physical servers from a primary site to a secondary site (often in a different Azure region).
- ASR provides regional DR and can also replicate data natively if the app supports it.
- Note that replication between regions is asynchronous, so some data loss may occur.

Multi-Region DR

- For robust DR, consider multi-region setups.

- Use a global load balancer (such as Azure Front Door or Traffic Manager) to distribute traffic across multiple Azure regions.
- This approach ensures high availability even if an entire region becomes unavailable.

High Availability

Azure Backup and Site Recovery

Azure Backup

Azure Backup is a robust data protection solution that simplifies backup management while saving costs. Here's what you need to know:

1. **Simplified Data Protection:**
   a. Azure Backup streamlines data protection for various workloads, including virtual machines (VMs), databases, and files.
   b. It offers application-aware backups, ensuring consistent snapshots of your data.
   c. You can easily configure backup policies and retention settings.

2. **Cost-Effective**:
   a. Azure Backup eliminates the need for on-premises backup infrastructure.
   b. Pay only for the storage consumed by your backups.
   c. It's a cloud-native solution, so you don't have to worry about hardware maintenance.

3. **Integration with On-Premises Solutions:**
   a. Extend your existing backup solution to Azure by seamlessly integrating Azure Backup.
   b. Protect on-premises data assets using a simple, secure, and cost-effective approach.
   c. Monitor and manage backups centrally through the Azure portal.

4. **Application Consistency:**
   a. Azure Backup ensures that backups are application-consistent, allowing for reliable restores.
   b. Whether it's a SQL database, a VM, or a file share, your data remains consistent during recovery.

Azure Site Recovery (ASR)

ASR provides disaster recovery protection for your Azure VMs and on-premises workloads. Here's what you need to know about ASR:

1. **Cloud-Native DR Solution:**
   a. ASR replicates VMs and physical servers from a primary site to a secondary site (often in a different Azure region).
   b. In cases of service disruption, accidental data deletion, or corruption, ASR orchestrates failover to the secondary site.

2. **High Churn Option:**
   a. ASR supports VMs with data churn up to 100 MB/s.
   b. This makes it suitable for IO-intensive workloads, ensuring disaster recovery for critical applications.

3. **Enterprise-Scale Disaster Recovery:**
   a. Host your servers in an on-premises data center and fail over to Azure infrastructure.
   b. Maintain a patched, supported, and highly-available environment during failover.

4. **Small and Midsize Business (SMB) DR:**
   a. SMBs can implement cost-effective disaster recovery using ASR or partner solutions.
   b. Azure Site Recovery enables inexpensive cloud-based DR for smaller organizations.

Azure Blob, Files: Backup & Recovery

Azure Blob Backup and Recovery

Operational Backup

- **Operational backup** protects your data using blob platform capabilities.
- Key features:
  - **Point-in-time restore**: You can restore blob data to an earlier state. This leverages soft delete, change feed, and blob versioning to retain data for a specified duration.
  - **Delete lock** prevents accidental or unauthorized deletion of the storage account. Operational backup automatically applies a delete lock to reduce the risk of data loss due to storage account deletion.

Vaulted Backup (Preview)

- **Vaulted backup** uses object replication to copy data to the backup vault.
- How it works:
    - Asynchronously copy block blobs between a source storage account and a destination storage account.
    - Copies blob contents, associated versions, metadata, and properties.
    - Configuration:
        - Azure Backup allocates a destination storage account (managed by Azure Backup) for storing backup data.
        - object replication policy is enabled at the container level on both the destination and source storage accounts.

Restoring Azure Blobs

To initiate a restore through the backup center, follow these steps:

1. In the Backup Center, go to **Restore** on the top bar.
2. On the **Initiate Restore** tab, choose **Azure Blobs (Azure Storage)** as the data source type.
3. Select the **backup instance** (storage account) containing the blobs you want to restore.
4. On the **Select Recovery Point** tab, choose the type of backup you want to restore

Azure VM, SQL - Backup & Recovery

Backup and Restore Options for SQL Server on Azure VMs

Automated Backup

- **SQL Versions**: 2014 and later
- **Description**:
    - **Automated Backup** allows you to schedule regular backups for all databases on a SQL Server VM.
    - Backups are stored in Azure storage for up to 30 days.
    - Starting with SQL Server 2016, Automated Backup offers additional options:
        - Configure manual scheduling.
        - Set the frequency to full and log backups.

Azure Backup for SQL VMs

- **SQL Versions**: 2012 and later
- **Description**:

    - **Azure Backup** provides an enterprise-class backup capability for SQL Server on Azure VMs.
    - You can centrally manage backups for multiple servers and thousands of databases.
    - Databases can be restored to a specific point in time using the Azure portal.
    - Offers customizable retention policies that can maintain backups for years.

Manual Backup

- **Applicable to All SQL Versions**
- **Description**:
    - Depending on your SQL Server version, there are various techniques for manually backing up and restoring SQL Server on Azure VMs.
    - In this scenario, you are responsible for:
        - How are your databases backed up?
        - The storage location and management of these backups.

Restoring SQL Server Databases on an Azure VM

To restore a SQL Server database running on an Azure VM backed up by Azure Backup, follow these steps:

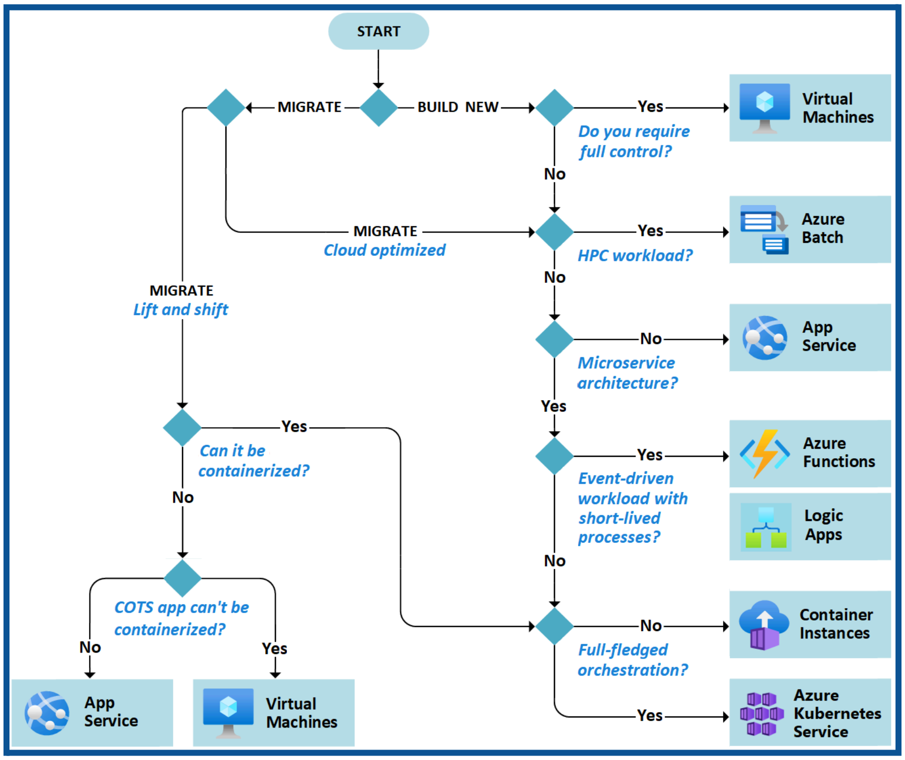
1. In the Azure portal, go to the **Backup Center** and click **Restore**.
2. Select **SQL in Azure VM** as the data source type.
3. Choose the database you want to restore and click **Continue**.
4. Specify where (or how) to restore the data:
   Alternate Location: Restore the database to an alternate location while keeping the source database.

Design Infrastructure Solutions

Design compute solutions

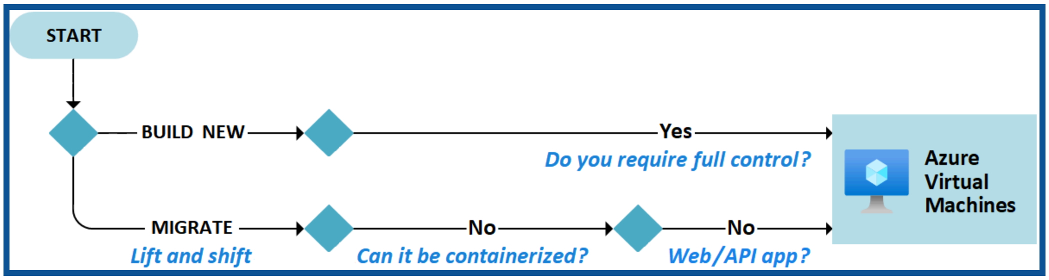
Azure Compute Services

Azure Compute Services are those services that are a fundamental part of Microsoft Azure, providing the infrastructure, tools, and platforms for deploying and managing workloads in the cloud.



Source: Microsoft Documentation

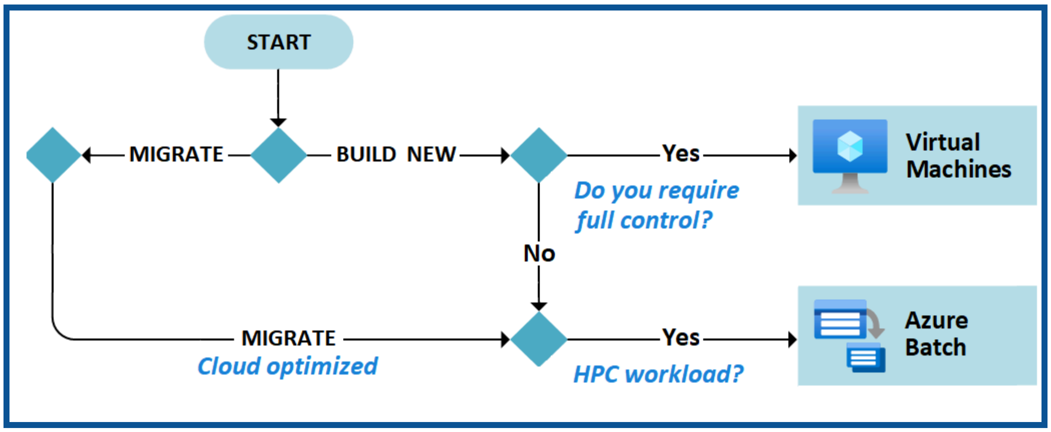
1. **Azure Virtual Machines (VMs):**
    - VMs allow you to deploy and manage virtual machines inside an Azure virtual network. You can choose from a variety of pre-configured VM sizes and operating systems.
    - Use cases: running legacy applications, hosting databases, or running custom software.



Source: Microsoft Documentation

2. **Azure App Service:**
   - A managed service for hosting web apps, mobile app backends, RESTful APIs, and automated business processes.
   - Provides automatic scaling, patching, and high availability.
   - Use cases: Web applications, APIs, and mobile backends.
   - Explore Azure App Service.
3. **Azure Functions:**
   - A serverless computing service that allows you to run event-driven code without managing infrastructure.
   - Ideal for small, stateless functions triggered by events.
   - Use cases: Event processing, data transformations, and automation.
   - Learn more about Azure Functions.
4. **Azure Kubernetes Service (AKS):**
   - A managed Kubernetes service for running containerized applications.
   - Simplifies container orchestration, scaling, and management.
   - Use cases: Containerized microservices, web applications, and AI/ML workloads.
   - Discover Azure Kubernetes Service.
5. **Azure Container Instances (ACI):**
   - A fast and simple way to run containers in Azure without managing VMs.
   - Ideal for short-lived tasks or single containers.
   - Use cases: Batch processing, testing, and development.
   - Explore Azure Container Instances.
6. **Azure Service Fabric:**
   - A distributed systems platform for building and managing scalable, reliable, and microservices-based applications.
   - Supports stateful and stateless services.
   - Use cases: Complex applications with high scalability requirements.
   - Learn more about Azure Service Fabric.

7. **Azure Batch:**
   - A managed service for running large-scale parallel and high-performance computing (HPC) applications.
   - Distributes workloads across a pool of VMs.
   - Use cases: Rendering, simulations, and scientific computing.
   - Discover Azure Batch.



Source: Microsoft Documentation

Azure Virtual Machines solutions & Batch solutions.

Azure Virtual Machines (VMs)

Azure Virtual Machines (VMs) are a fundamental computing resource in Microsoft Azure. Here are some key points about VMs:

1. Purpose:
   - VMs allow you to create and manage virtualized computing environments in the cloud.
   - You can choose from a variety of pre-configured VM sizes and operating systems, including both Linux and Windows1.
2. Use Cases:
   - Running legacy applications that require specific OS versions.
   - Hosting databases, web servers, or custom software.
   - Development and testing environments.
   - High-performance computing (HPC) workloads.
3. Cost Optimization:
   - Use Azure Reserved Virtual Machine Instances to save costs.
   - Reuse on-premises licenses with Azure Hybrid Benefit for Windows Server.

4. Scalability:
    - Scale VMs up or down based on workload requirements.
    - Use Azure Spot Virtual Machines for cost-effective workloads.
5. Management:
    - Manage VMs through the Azure portal, CLI, or APIs.
    - Monitor performance, apply patches, and configure networking.

Learn more about Azure Virtual Machines and explore the available VM series.

Azure Batch

Azure Batch is designed for running large-scale parallel and high-performance computing (HPC) workloads efficiently in the cloud. Here's what you need to know:

1. How It Works:
    - Azure Batch creates and manages a pool of compute nodes (virtual machines).
    - You install the applications you want to run and schedule jobs to execute on these nodes.
    - No need to install or manage cluster or job scheduler software.
2. Workload Types:
    - **Intrinsically Parallel Workloads**:
        - These workloads can run independently, completing parts of the work.
        - Examples: Financial risk modeling, image rendering, data processing, and software testing.
    - **Tightly Coupled Workloads**:
        - Applications that need to communicate with each other (use MPI).
        - Examples: Finite element analysis, fluid dynamics, and multi-node AI training.
3. Additional Capabilities:
    - Azure Batch supports rendering tools (e.g., Autodesk Maya, 3ds Max) and integrates with Azure Data Factory for data transformation.
    - Optimize performance using specialized HPC and GPU-optimized VM sizes.

Azure App Service and Function Solutions

Azure App Service Solutions

Azure App Service is a platform-as-a-service (PaaS) offering provided by Microsoft Azure. It enables developers to quickly build, deploy, and scale web, mobile, and API applications without worrying about the underlying infrastructure. Here are the key points:

1. Purpose:
    - Azure App Service allows you to build and host:
        - **Web Apps**: For creating web applications.
        - **Mobile Back Ends**: For mobile app back-end services.
        - **RESTful APIs**: For building APIs that follow REST principles.
    - You can develop in your favorite language, including **.NET**, **.NET Core**, **Java**, **Node.js**, **PHP**, and **Python**.
2. Features:
    - **Auto-Scaling and High Availability**:
        - Apps automatically scale based on demand.
        - Offers a service-level agreement (SLA)-backed uptime of 99.95 percent.
    - **Cross-Platform Support**:
        - Applications run and scale with ease on both **Windows** and **Linux**-based environments.
    - **Automated Deployments**:
        - Deploy directly from **GitHub**, **Azure DevOps**, or any **Git** repository.
    - **Built-in Infrastructure Maintenance**:
        - No need to manage servers or patching.
    - **Security and Compliance**:
        - Rigorous standards, including **SOC** and **PCI**, for seamless deployments.
        - Use **Azure Web Application Firewall** for additional protection.
        - Authenticate and authorize app access using **Azure Active Directory** and other identity providers.
3. Developer Productivity:
    - Tight integration with **Visual Studio Code** and **Visual Studio**.
    - Streamline **CI/CD** with tools like **Git**, **GitHub**, **GitHub Actions**, **Azure DevOps**, **Docker Hub**, and **Azure Container Registry**.
    - Reduce downtime and minimize risk during app updates using **deployment slots**.
4. Cost Efficiency:
    - Focus on app innovation rather than managing infrastructure.

- A commissioned study found that **Azure PaaS delivers 228 percent ROI over 3 years**, with a 15-month payback period

Azure Container Instances & Kubernetes Service solutions

Azure Container Instances (ACI)

Azure Container Instances (ACI) is a managed serverless compute service that allows you to run Docker containers in a simplified and isolated environment.

Here are some key points about ACI:

1. Serverless and Isolated:
    - ACI abstracts away the underlying infrastructure, so you can focus solely on your application code.
    - There is no need to manage virtual machines (VMs) or orchestration tools.
    - Ideal for scenarios where you need to run containers without the complexity of Kubernetes1.
2. Use Cases:
    - **Elastic Bursting with AKS**:
        - ACI provides fast, isolated computing to handle traffic spikes without managing servers.
        - Azure Kubernetes Service (AKS) can use the Virtual Kubelet to provision pods inside ACI, scaling out as needed.
    - **Event-Driven Applications**:
        - Combine ACI with Azure Logic Apps, Azure queues, and Azure Functions to build robust infrastructure that scales out containers on demand.
    - **Data Processing Jobs**:
        - Use ACI for data processing tasks, achieving cost savings through per-second billing compared to statically-provisioned VMs.
3. Security and Compliance:
    - ACI provides hypervisor isolation for each container group, ensuring containers run in isolation without sharing a kernel.
    - Microsoft invests heavily in cybersecurity research and development, with over 3,500 security experts dedicated to data security and privacy2.

Key Benefits

- **Fast Startup Times**: Containers in ACI start up quickly, providing agility for your workloads.

- **Container Access**: ACI allows exposing container groups directly to the internet with an IP address and a fully qualified domain name (FQDN).
- **Compliant Deployments**: Use Azure Policy for regulatory compliance controls.
- **Custom Sizes**: Define resources (vCPU and memory) per container group.
- **Linux and Windows Containers**: ACI supports both Linux and Windows containers.
- **Co-Scheduled Groups**: Run multiple containers together in the same group.
- **Virtual Network Deployment**: Deploy ACI containers within a virtual network for additional security.

Azure Kubernetes Service (AKS)

Azure Kubernetes Service (AKS) simplifies deploying and managing Kubernetes clusters in Azure. Here's what you need to know:

1. Managed Kubernetes Cluster:
   - AKS provides a fully managed Kubernetes environment.
   - Azure handles critical tasks like health monitoring and maintenance.
   - When you create an AKS cluster, a control plane is automatically set up and configured4.
2. Developer Productivity:
   - Debug microservice applications using Kubernetes extensions for Visual Studio and Visual Studio Code.
   - Add CI/CD pipelines through GitHub Actions and simplify runtime with Dapr.
   - Gain observability into your environment using Kubernetes resource views and telemetry.
3. Security and Compliance:
   - Enforce regulatory compliance controls using Azure Policy.
   - Fine-grained identity and access control with Azure Active Directory.
   - Use Microsoft Defender for Containers for improved security.
4. Flexible Deployment Options:
   - Deploy AKS on the infrastructure of your choice using Azure Arc, from the cloud to the edge2.

Key Benefits

- **Unified Management**: AKS offers unified management and governance for on-premises, edge, and multi-cloud Kubernetes clusters.
- **End-to-end Developer Productivity**: Debugging, CI/CD, logging, and automated node maintenance.

- **Advanced Identity and Access Management**: Monitor and maintain container security at scale.
- **Trusted Platform**: AKS integrates seamlessly with other Azure services, making it easy to deploy and manage applications with Azure DevOps, Azure Container Registry, and more

Azure Functions solutions

Azure Functions

Azure Functions is a serverless solution that allows you to write less code, maintain less infrastructure, and save on costs. Instead of worrying about deploying and managing servers, Azure Functions provides all the up-to-date resources needed to keep your applications running. Here are some key points about Azure Functions:

1. **Scenarios:**
    - **Process File Uploads**:
        - Run code when a file is uploaded or changed in blob storage.
    - **Real-Time Data Processing**:
        - Capture and transform data from event and IoT source streams on the way to storage.
    - **Machine Learning Workflows**:
        - Build machine learning workflows with a serverless architecture.
    - **Serverless APIs**:
        - Create APIs using HTTP triggers with Node.js or Microsoft. NET.
    - **Serverless Web Applications**:
        - Host static websites and single-page applications.
    - **Serverless Microservices**:
        - Implement event-driven microservices.
    - **Data Processing Pipelines**:
        - Handle files and real-time data processing1.

2. **Development Lifecycle:**
    - Write your function code in your preferred language (C#, Java, JavaScript, PowerShell, Python, and more).
    - Use popular development tools like Visual Studio, Visual Studio Code, and Maven for seamless debugging and deployments.
    - Integrate with Azure Monitor and Azure Application Insights for comprehensive runtime telemetry and analysis2.

3. **Hosting Options:**
    - Choose from various hosting options based on your business needs and workload:
        - **Consumption Plan**: Fully serverless, pay only for execution time.

- **Premium Plan**: Always warm instances for the fastest response times.

Azure Logic Apps solutions

Azure Logic Apps

Azure Logic Apps is a cloud platform where you can create and run automated workflows with little to no code. By using the visual designer and selecting from prebuilt operations, you can quickly build a workflow that integrates and manages your apps, data, services, and systems. Here are some key points about Azure Logic Apps:

1. Purpose and Use Cases:
   - **Automated Workflows**: Logic Apps simplify the way you connect legacy, modern, and cutting-edge systems across cloud, on-premises, and hybrid environments.
   - **Low-Code/No-Code Development**: You can develop highly scalable integration solutions supporting enterprise and business-to-business (B2B) scenarios.
   - **Example Scenarios**:
     - Schedule and send email notifications using Office 365 when specific events occur (e.g., a new file is uploaded).
     - Route and process customer orders across on-premises systems and cloud services.
     - Move uploaded files from an SFTP or FTP server to Azure Storage.
     - Monitor tweets, analyze sentiment, and create alerts or tasks for review.
2. Core Terminology and Concepts:
   - **Logic App**: The Azure resource you create when building a workflow. You can create two types of logic app resources:
     - **Consumption Logic App**: Supports a single workflow, hosted and run in global multitenant Azure Logic Apps.
     - **Standard Logic App**: Supports multiple workflows, hosted and run in single-tenant Azure Logic Apps.
   - **Workflow**: A series of operations defining a task, business process, or workload. It starts with a trigger operation followed by one or more action operations.
   - **Trigger**: The first operation in any workflow, specifying criteria before running subsequent operations (e.g., getting an email or detecting a new file).
3. Benefits:
   - **Integration Platform as a Service (iPaaS)**: Logic Apps is a leading iPaaS built on a containerized runtime, allowing deployment and execution anywhere.
   - **Scale and Portability**: Deploy and run Logic Apps anywhere, automating business-critical workflows.
   - **Developer-Friendly**: Use low-code/no-code tools for rapid development.

- **Common Scenarios**: Especially useful for coordinating actions across multiple systems and services.

Design an application architecture

Azure Queue Storage and Azure Service Bus

Azure Queue Storage

1. Purpose:
    - **Azure Queue Storage** is part of the Azure Storage infrastructure.
    - It allows you to store large numbers of messages.
    - You can access messages from anywhere in the world via authenticated calls using HTTP or HTTPS.
    - A queue message can be up to 64 KB in size.
    - Queues are commonly used to create a backlog of work to process asynchronously.
2. Scalability and Simplicity:
    - Designed for high-throughput scenarios that require a large volume of messages.
    - Ideal for scenarios where you need a simple, cost-effective message queue.
3. Use Cases:
    - Tracking progress for processing messages.
    - Storing over 80 gigabytes of messages in a queue.
    - Server-side logs of all transactions executed against your queues.

Azure Service Bus

1. Purpose:
    - **Azure Service Bus** is part of a broader Azure messaging infrastructure.
    - It supports queuing, publish/subscribe, and more advanced integration patterns.
    - Designed to integrate applications or components across different communication protocols, data contracts, trust domains, or network environments.
2. Advanced Capabilities:
    - Offers guaranteed first-in-first-out (FIFO) ordered delivery.
    - Supports automatic duplicate detection.
    - Allows long-polling receive operations without polling the queue.
    - Enables processing messages as parallel long-running streams.
3. Use Cases:
    - High-value enterprise messaging scenarios.
    - Complex integration patterns.
    - Guaranteed message order and duplicate detection.

Azure Event Hubs and Event Grid

Azure Event Hubs

Azure Event Hubs is a fully managed, real-time data ingestion service that allows you to stream millions of events per second from any source.

Here are some key points about Event Hubs:

1. Purpose:
   - Event Hubs is designed for high-volume telemetry streaming.
   - It ingests and processes large volumes of events and data with low latency and high reliability.
   - Ideal for scenarios where you need to collect and process massive amounts of data from distributed software and devices.
2. Features:
   - **MQTT Messaging**:
     - Event Hubs supports MQTT v3.1.1 and v5.0 protocols, allowing IoT devices and applications to communicate efficiently.
   - **Custom Topics with Wildcards**:
     - Create your topic structure and leverage wildcards for flexible event routing.
   - **Publish-Subscribe Model**:
     - Efficiently communicate using one-to-many, many-to-one, and one-to-one messaging patterns.
   - **Built-in Cloud Integration**:
     - Route MQTT messages to Azure services or custom webhooks for further processing.
   - **Scalability**:
     - Adjust throughput dynamically based on usage needs and pay only for what you use.
3. Use Cases:
   - Ingesting data from IoT devices.
   - Building real-time big data pipelines.
   - Integrating with Azure services for further analysis, visualization, or storage.

Azure Event Grid

Azure Event Grid is a highly scalable, fully managed Pub/Sub message distribution service.

Here are the key points about Event Grid:

1. Purpose:
   - Event Grid enables you to build event-driven serverless architectures.
   - It allows clients to publish and subscribe to messages over MQTT and HTTP protocols.
   - Supports Internet of Things (IoT) solutions through MQTT.
   - Provides push and pull delivery modes for data distribution.
2. Features:
   - **MQTT Messaging**:
     - Event Grid supports MQTT v3.1.1 and v5.0, making it suitable for IoT scenarios.
   - **Push and Pull Delivery**:
     - Send events to subscribers (push delivery) or allow subscribers to read events (pull delivery).
   - **Built-in Cloud Integration**:
     - Route events to Azure services or custom webhooks.
   - **CloudEvents 1.0 Support**:
     - Ensures interoperability across systems.
3. Use Cases:
   - Integrating applications.
   - Building reactive scenarios (e.g., item shipped notifications).
   - Creating event-driven solutions.

Azure API Management

Azure API Management: An Overview

Azure API Management is a cloud-based service offered by Microsoft Azure that allows businesses to create, publish, and manage APIs. It provides a centralized platform for exposing APIs to internal and external developers, partners, and customers. Here are some key points about Azure API Management:

1. API Lifecycle Management:
   - Azure API Management supports the complete API lifecycle, from creation to retirement.
   - It helps organizations manage APIs as first-class assets throughout their lifecycle.
2. Common Scenarios:
   - **Unlocking Legacy Assets**:

        - APIs abstract and modernize legacy backends, making them accessible from new cloud services and modern applications.
        - APIs allow innovation without the risk, cost, and delays of migration.
    - **API-Centric App Integration**:
        - APIs simplify and reduce the cost of application integration by providing easily consumable, standards-based mechanisms for exposing and accessing data, applications, and processes.
    - **Multi-Channel User Experiences**:
        - APIs enable user experiences across web, mobile, wearable, or Internet of Things applications.
        - Reusing APIs accelerates development and return on investment.
    - **B2B Integration**:
        - Exposing APIs to partners and customers lowers the barrier to integrating business processes and exchanging data.
        - APIs eliminate the overhead inherent in point-to-point integration and scale B2B integration with self-service discovery and onboarding.
3. Components:
    - **API Gateway**:
        - Acts as a facade to backend services, allowing API providers to abstract API implementations.
        - Enables consistent configuration of routing, security, throttling, caching, and observability.
    - **Management Plane**:
        - Provides tools for creating, publishing, and managing APIs.
        - Includes features like authentication, authorization, usage limits, mocking, documentation, and more.
    - **Developer Portal**:
        - A self-service portal for developers to discover, explore, and consume APIs.
4. Integration with Azure Services:
    - Azure API Management integrates with complementary Azure services, such as Azure Key Vault for secure management of certificates and secrets, and Azure Monitor for logging, reporting, and alerting on management operations and API requests.

Azure Cache for Redis

Azure Cache for Redis is a fully managed, distributed, in-memory caching solution provided by Microsoft Azure. It enables you to create a caching layer for your applications, improving performance, reducing latency, and handling high-traffic loads.

Here are some key points about Azure Cache for Redis:

1. Purpose and Use Cases:
    - **Caching Layer**:
        - Use Azure Cache for Redis as a distributed data or content cache.
        - Improve application throughput and latency by storing frequently accessed data in memory.
    - **Session Store**:
        - Efficiently store and manage session data (e.g., user cookies, output pages) for web applications.
        - Enhance application responsiveness and handle increasing loads with fewer compute resources.
    - **Message Broker**:
        - Implement publish/subscribe or queue architectures using Azure Cache for Redis.
        - Route real-time messages and scale up communication frameworks (e.g., SignalR).
    - **Integration with Other Azure Services**:
        - Deploy standalone or alongside other Azure database services (e.g., Azure SQL, Azure Cosmos DB).
2. Features:
    - **High Throughput and Low Latency**:
        - Achieve superior throughput, handling millions of requests per second with sub-millisecond latency.
    - **Fully Managed Service**:
        - Automatic patching, updates, scaling, and provisioning.
    - **Redis Data Structures**:
        - Natively supports Redis data structures like hashes, lists, and sorted sets.
    - **Module Integration**:
        - Supports RedisBloom, RediSearch, RedisJSON, and RedisTimeSeries modules for data analysis, search, and streaming.
    - **Powerful Capabilities**:
        - Clustering, active geo-replication, Redis on Flash, and up to 99.999% availability.

3. Redis Versions Supported:
   ○ Azure Cache for Redis offers both the open-source Redis (OSS Redis) and Redis Enterprise (commercial product from Redis Inc.) as a managed service.
   ○ Currently supports OSS Redis versions 4.0.x and 6.0.x.
   ○ You can choose from newer major releases and at least one older stable version

Azure Automation Servrices [ARM and Bicep Files]

Azure Resource Manager (ARM) templates

Azure Resource Manager (ARM) templates are JavaScript Object Notation (JSON) files that define the infrastructure and configuration for your project in Microsoft Azure. These templates allow you to declare the desired state of resources, their dependencies, and their properties. Here are some key points about ARM templates:

1. **What Are ARM Templates?**
   ○ ARM templates provide a structured approach to managing resources in Azure.
   ○ They allow you to define the objects you want (such as virtual machines, storage accounts, or networks) and their properties in a declarative JSON file.
   ○ Think of ARM templates as “Infrastructure as Code” – a way to define and deploy your Azure resources consistently and reliably.
2. **Use Cases and Benefits:**
   ○ **Infrastructure Deployment**:
      ■ ARM templates enable you to define the infrastructure requirements for your deployments on Azure.
      ■ You treat ARM templates like application code, checking them into source control and managing them just like any other code file.
   ○ **Consistency and Repeatability**:
      ■ By using templates, you ensure that your resources are consistently deployed across different environments (development, staging, production).
      ■ You can easily replicate your infrastructure by deploying the same template multiple times.
   ○ **Version Control and Collaboration**:
      ■ Store your templates in version control systems (such as Git) for collaboration and change tracking.
      ■ Share templates with your team, making it easier to manage and maintain your infrastructure.
3. **Components of an ARM Template:**
   ○ **Parameters**: Define input values that can be customized during deployment.

- **Variables**: Store intermediate values or reusable expressions.
- **Resources**: Specify the Azure resources you want to create or configure.
- **Outputs**: Define values that are returned after deployment (e.g., connection strings, IP addresses).

4. **Deployment Modes:**
   - **Incremental Deployment**: Updates or adds resources without affecting existing resources.
   - **Complete Deployment**: Deletes and recreates the entire resource group, ensuring a clean slate.
   - **Validation Deployment**: Checks the template syntax and validates resource references without actually deploying resources.
5. **Deployment Methods:**
   - **Azure Portal**: Create and deploy templates directly from the Azure portal.
   - **Visual Studio Code (VS Code)**: Use extensions like "Azure Resource Manager Tools" to create and edit templates.
   - **PowerShell**, **Azure CLI**, and **REST API**: Automate deployments using command-line tools.
   - **Azure DevOps**: Integrate ARM templates into your CI/CD pipelines.

Azure Bicep templates and Azure Automation

Azure Bicep Templates

Azure Bicep is a domain-specific language (DSL) that simplifies the process of defining and deploying Azure resources. It is an abstraction layer over ARM templates, making writing and maintaining infrastructure-as-code (IaC) easier.
Here are some key points about Bicep:

1. **Advantages of Bicep:**
   - **Improved Syntax**: Bicep offers a cleaner and more concise syntax than ARM templates.
   - **Readability**: Bicep files are easier to read and understand, reducing the cognitive load for developers.
   - **Type Safety**: Bicep provides strong typing and validation during development.
   - **Transpiration**: Bicep files are transpiled into ARM template JSON during deployment.
2. **Creating Bicep Templates:**
   - Use Visual Studio Code (VS Code) with the Bicep extension to create Bicep files.
   - Leverage Intellisense for template properties by adding the Azure Resource Manager Tools extension.
3. **Resource Deployment:**
   - Bicep templates define the desired state of Azure resources.
   - Deploy Bicep files using Azure CLI, Azure PowerShell, or Azure DevOps pipelines.

Azure Automation

Azure Automation is a cloud-based service that allows you to automate various tasks related to managing and maintaining your Azure and non-Azure environments. Here's what you need to know:

1. **Process Automation:**
   - Azure Automation helps automate frequent, time-consuming, and error-prone management tasks.
   - Use graphical, PowerShell, or Python runbooks to create automation workflows.
   - Hybrid Runbook Workers allow running runbooks against on-premises or other cloud resources.
2. **Configuration Management:**
   - Change Tracking and Inventory: Track changes across Linux and Windows virtual machines and servers.
   - Azure Automation State Configuration: Define and enforce desired configurations for your resources.
3. **Webhooks**:
   - Fulfill requests and trigger automation from Azure Logic Apps, Azure Functions, ITSM tools, DevOps pipelines, and monitoring systems.
4. **Benefits:**
   - Lower operational costs by reducing errors and boosting efficiency.
   - Consistent management across Azure and non-Azure environments.
   - Integration with other Azure services and third-party systems.

Azure App Configuration

Azure App Configuration: An Overview

Azure App Configuration is a service that allows developers to centrally manage application settings and feature flags. It provides a way to store configuration settings and securely access them from various components of your application. Here are some key points about Azure App Configuration:

1. Why use Azure App Configuration?
   - **Centralized Configuration**:
     - Modern cloud-based applications often consist of distributed components running on multiple virtual machines, containers, and services.
     - Spreading configuration settings across these components can lead to errors during deployment and troubleshooting.

        - Azure App Configuration centralizes all your application settings in one place.
    - **Separation of Configuration from Code**:
        - Following best practices (such as the Twelve-Factor App), it's recommended to keep configuration separate from code.
        - Externalize configuration settings to an external source or runtime environment.
    - **Scalable and secure**:
        - Azure App Configuration is fully managed and can scale to meet your application's needs.
        - It provides enhanced security through Azure-managed identities and encryption of sensitive information.
2. Use Cases:
    - **Microservices and Containers**:
        - Azure Kubernetes Service (AKS), Azure Service Fabric, and other containerized apps benefit from centralized configuration management.
    - **Serverless Apps**:
        - Azure Functions and other event-driven stateless compute apps can dynamically change settings without redeployment.
    - **Continuous Deployment Pipelines**:
        - Manage configuration settings across different environments (dev, test, production) without code changes.
    - **Feature Flags**:
        - Control feature availability in real-time using Azure App Configuration's dedicated UI for feature flag management.
3. Key Features:
    - **Flexible Key-Value Store**:
        - Store configuration settings as key-value pairs.
        - Use labels for versioning and organization.
    - **Point-in-Time Replay**:
        - Roll back to previous configurations if needed.
    - **Enhanced Security**:
        - Azure-managed identities and encryption at rest and in transit.
    - **Integration with Popular Frameworks**:
        - Native integration with popular frameworks like .NET, Java, Python, and Node.js.
4. Comparison with Azure Key Vault:
    - While Azure Key Vault is used for storing secrets (such as passwords, certificates, and API keys), Azure App Configuration focuses on configuration settings.
    - App Configuration complements Key Vault by making it easier to manage hierarchical configuration data.

Design migrations

Evaluate migration with the Cloud Adoption Framework

When considering migrating workloads to the cloud, it's essential to assess their readiness. Here are some steps and best practices for evaluating workload readiness:

Source: Microsoft Documentation

1. Evaluation Assumptions:
    - The readiness evaluation process is specific to each cloud platform. For our discussion, let's assume an intention to migrate to **Azure**.
    - We'll use **Azure Migrate** (also known as Azure Site Recovery) for replication activities. However, alternative tools are available.

2. Common Infrastructure Evaluation Activities:
    - **VMware Requirements**: Review the Azure Site Recovery requirements for VMware.
    - **Hyper-V Requirements**: Review the Azure Site Recovery requirements for Hyper-V.
3. Common Database Evaluation Activities:
    - **Recovery Point Objectives (RPOs) and Recovery Time Objectives (RTOs)**: Document these for the current database deployment. They aid in decision-making during architecture activities.
    - **High-Availability Configuration**: Document any requirements for high availability.
    - **PaaS Compatibility Evaluation**: Map on-premises databases to compatible Azure PaaS solutions (e.g., Azure Cosmos DB, Azure SQL Database, Azure Database for MySQL, PostgreSQL, or MariaDB). Consider PaaS migrations for time savings and reduced total cost of ownership (TCO).
    - If PaaS compatibility requires remediation, consult the teams responsible for architecture and remediation activities.

Describe the Azure migration framework

Azure Migration Framework based on the AZ-305 revision purpose. As an Azure solutions architect, understanding migration frameworks is crucial for designing effective solutions. Here's what you need to know:

1. Purpose of the Azure Migration Framework:
    - The **Azure Migration Framework** provides a structured approach to planning and executing cloud migrations.
    - It helps organizations assess, prioritize, and migrate workloads to Azure efficiently.
    - The framework aligns with best practices, industry standards, and Microsoft's expertise.

2. Components of the Azure Migration Framework:
    - **Assessment Phase**:
        - Evaluate existing on-premises servers, data, and applications.
        - Leverage tools like **Azure Migrate** to assess readiness and identify dependencies.

    - **Migration Solution Design**:
        - Recommend migration solutions based on workload characteristics.
        - Consider infrastructure as a service (IaaS) and platform as a service (PaaS) options.
    - **Execution and Optimization**:
        - Execute the migration plan, ensuring minimal downtime.
        - Optimize resources post-migration for cost-effectiveness and performance.

3. Microsoft Cloud Adoption Framework (CAF):
    - The **CAF** provides a comprehensive guide for cloud adoption.
    - It covers strategy, planning, readiness, and governance.
    - The **Azure Migration Framework** aligns with the **CAF** principles.

4. Tools and Resources:
    - **Azure Migrate**: Assess, compare, and migrate workloads.
    - **Azure Site Recovery**: Replicate and fail over VMs.
    - **Azure Database Migration Service**: Migrate databases.
    - **Azure Data Box**: Transfer large data sets offline.
    - **Azure Cost Management and Billing**: Monitor and optimize costs.

Assess your on-premises workloads

Assessing your on-premises workloads and understanding workload assessment is crucial for designing effective migration strategies.
Here are the key steps:

1. Evaluate On-Premises Servers, Data, and Applications:
    - Begin by assessing your existing on-premises infrastructure. Understand the servers, applications, and data that need migration.
    - Consider factors like performance, dependencies, security, and compliance.

2. Select a Migration Tool:
    - Choose the right tool for assessing and migrating workloads. Some popular options include:
        - **Azure Migrate**: Assess readiness and identify dependencies.
        - **Azure Site Recovery**: Replicate and fail over VMs.
        - **Azure Database Migration Service**: Migrate databases.

3. Recommend a Solution for Migrating Workloads:
    - Based on your assessment, recommend the appropriate migration approach:
        - **Infrastructure as a Service (IaaS)**: Lift and shift VMs to Azure.
        - **Platform as a Service (PaaS)**: Consider migrating to Azure PaaS solutions (e.g., Azure SQL Database, Cosmos DB) for scalability and reduced TCO.

4. Consider Pricing Tiers and Geographic Scopes:
    - Understand the differences in pricing tiers for Azure services.
    - Be aware of geographic service availability (e.g., Azure AD P2, Premium SKUs).

Azure Migrate Overview

Azure Migrate provides a simplified migration, modernization, and optimization service for Azure. It encompasses all pre-migration steps, including discovery, assessments, and right-sizing of on-premises resources related to infrastructure, data, and applications.

Here are the key points:

1. Unified Migration Platform:
- **Azure Migrate** offers a single portal to initiate, manage, and track your migration journey to Azure.
- It streamlines the migration process by providing a centralized hub for all migration-related activities.

2. Range of Tools:
- **Azure Migrate** includes various tools for assessment and migration:

    - **Azure Migrate: Discovery and Assessment**: This tool helps discover and assess on-premises servers, including SQL and web apps. It supports VMware, Hyper-V, and physical servers, preparing them for migration to Azure.
    - **Migration and Modernization**: Use this tool to migrate servers, including VMware VMs, Hyper-V VMs, physical servers, other virtualized servers, and public cloud VMs to Azure.
    - **Data Migration Assistant**: Assess SQL Server databases for migration to Azure SQL Database, Azure SQL Managed Instance, or Azure VMs running SQL Server. It identifies potential issues blocking migration and recommends the right migration path.
    - **Azure Database Migration Service**: Migrate on-premises databases to Azure VMs running SQL Server, Azure SQL Database, or SQL Managed Instances.
    - **Movere**: Assess servers for migration.

3. Assessment, Migration, and Modernization:
    - **Servers**: Assess on-premises servers (including web apps and SQL Server instances) and migrate them to Azure.
    - **Databases**: Assess on-premises SQL Server instances and databases, migrating them to Azure SQL Database, Azure SQL Managed Instance, or an Azure VM running SQL Server.
    - **Web Applications**: Assess on-premises web applications and migrate them to Azure App Service and Azure Kubernetes Service.
    - **Virtual Desktops**: Assess on-premises virtual desktop infrastructure (VDI) and migrate it to Azure Virtual Desktop.
    - **Data**: Migrate large data volumes to Azure efficiently using Azure Data Box products.

Designing a Migration Solution

When designing a migration solution, consider the following steps:

1. **Discover and Assess:**
    - Use the **Azure Migrate: Discovery and Assessment** tool to understand your digital estate. Identify on-premises infrastructure, applications, and dependencies.
    - Assess servers, databases, and web apps to determine their readiness for migration.
2. **Plan Your Move:**
    - Leverage technical and business insights gained during assessment to plan your migration.
    - Consider factors like infrastructure requirements, database compatibility, and application dependencies.
3. **Migrate in Phases:**
    - Break down your migration into manageable phases.
    - Prioritize workloads based on criticality, complexity, and business impact.
4. **Modernize for Innovation:**
    - As you migrate, explore opportunities for modernization.
    - Consider Azure services like Azure App Service, Azure SQL Database, and Azure Kubernetes Service.

Migrate your structured data in databases

Migrating structured data in databases includes understanding database migration strategies which is crucial for designing effective solutions.
Here are the key steps and considerations:

1. **Assess Your Databases:**
   - Begin by evaluating your existing on-premises databases. Understand their schema, data types, and dependencies.
   - Consider factors like data volume, performance requirements, and security constraints.

2. **Choose the Right Migration Approach:**
   - **Lift and Shift (IaaS)**: Migrate your databases to Azure Virtual Machines (VMs) running SQL Server. This approach maintains compatibility but requires VM management.
   - **Platform as a Service (PaaS)**: Consider migrating to Azure SQL Database or Azure SQL Managed Instance. PaaS solutions offer scalability, automatic backups, and reduced administrative overhead.

3. **Data Migration Tools:**
   - **Azure Database Migration Service**: Use this service to migrate on-premises SQL Server databases to Azure SQL Database or Azure SQL Managed Instance.
   - **Data Migration Assistant (DMA)**: Assess and migrate databases. DMA helps identify compatibility issues and provides recommendations for remediation.

4. **Schema and Data Transformation:**
   - Ensure that your database schema is compatible with the target platform.
   - Address any data type mismatches, collation differences, and other schema-related issues.

5. **Security and Compliance:**
   - Review security requirements. Azure SQL Database offers features like Transparent Data Encryption (TDE), Always Encrypted, and Auditing.
   - Ensure compliance with industry standards (e.g., GDPR, HIPAA).

6. **High Availability and Disaster Recovery:**
   - Configure high availability options such as failover groups or geo-replication.
   - Set up automated backups and retention policies.

7. **Testing and Validation:**
   - Perform test migrations to validate the process and identify any issues.
   - Test application connectivity and functionality against the migrated database.

Azure Storage Migration for Unstructured Data

When migrating unstructured data (such as files and objects) to Azure, you have several options. Let's focus on the tools and considerations for this specific scenario:

Azure Migrate:

Purpose: Azure Migrate provides a centralized hub to assess and migrate on-premises servers, infrastructure, applications, and data to Azure.

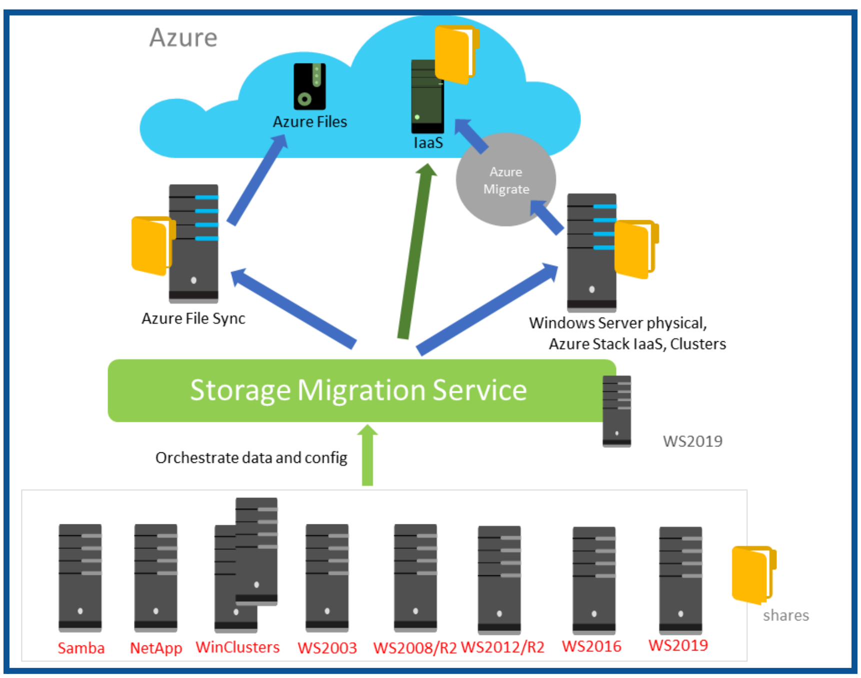
1. Unstructured Data Migration:

File Migration:

- From network-attached storage (NAS) to Azure file offerings:
- Azure Files: Managed file shares in the cloud.
- Azure NetApp Files: High-performance file storage powered by NetApp.
- Independent Software Vendor (ISV) Solutions: Third-party tools that facilitate file migration.

Object Migration:

- From object storage solutions to the Azure object storage platform:
- Azure Blob Storage: Scalable, durable object storage.
- Azure Data Lake Storage: Optimized for big data analytics.



Source: Microsoft Documentation

2. Assessment and Migration Phases

- **Discovery Phase**:
  - Identify sources to be migrated (e.g., SMB shares, NFS exports, or object namespaces).
  - Use automated tools or manual assessment.

- **Assessment Phase**:
  - Understand available migration options.
  - Consider technical and cost factors.
  - Choose a target storage service:
    - Evaluate aspects like protocol support, performance characteristics, and service limits.
    - Decision tree: Native Azure services → ISV solutions if needed.
  - Perform cost assessment for the most cost-effective option.

3. Migration Tools:

- **Commercial Tools (Azure and ISV)**:
  - Azure provides built-in tools for assessment and migration.
  - Independent software vendor (ISV) solutions offer additional features.
- **Open Source Tools**:
  - Explore community-supported tools for specific scenarios.